Story of How POSTECH Defeats KAIST

Mathematics for Science Quiz

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POSTECH–KAIST Science War

Pohang University of Science and Technology
vs
Korea Advanced Institute of Science and Technology

Preface

This book is devoted to the Mathematics section of Science Quiz in the POSTECH–KAIST Science War. It collects mathematical concepts, facts to memorize, past problems, and expected problems that are useful for competition preparation.

The main purpose of this text is not merely to record solutions, but to build a practical and reliable mathematical toolkit for Science Quiz. It includes standard theorems, important terminology, memorable facts about mathematicians and mathematical history, and problem-solving strategies that may be useful in an actual match.

I sincerely hope that this book contributes to POSTECH's victory. Good luck, and let the game begin.

Contents

Notation and Conventions

Throughout this book, we use standard mathematical notation whenever possible. The conventions below are arranged to keep the exposition readable, unified, and consistent with the chapter structure of Part I. The guiding principle is simple: use official, standard notation first, and introduce special notation only when it makes a quiz solution faster or clearer.

General Conventions

Scalars are written in ordinary italic letters, such as a, b, c, λ, t. Vectors are written in bold lowercase letters, such as $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Matrices are written in bold uppercase letters, such as $\mathbf{A}, \mathbf{B}, \mathbf{P}$. Unless otherwise stated, vectors are regarded as column vectors. Named theorems, standard English terminology, and conventional symbols are preferred over ad hoc notation.

Notation	Meaning
$\mathbb{N}$	the set of positive integers, $\{1, 2, 3, \dots\}$
$\mathbb{N}_0$	the set of nonnegative integers, $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	the set of integers
$\mathbb{Q}$	the set of rational numbers
$\mathbb{R}$	the set of real numbers
$\mathbb{C}$	the set of complex numbers
$A \subseteq B$	A is a subset of B
$A \subsetneq B$	A is a proper subset of B
$A \setminus B$	the set difference of A and B
$\lvert A \rvert$	the cardinality of a finite set A
$\varnothing$	the empty set
λ	usually a scalar, eigenvalue, or parameter, depending on context
ε	a small positive real number, especially in analysis
$\operatorname{Re} z$	the real part of a complex number z
$\operatorname{Im} z$	the imaginary part of a complex number z
$\sim$	may mean asymptotic equivalence, distribution, or an equivalence relation
$O(\cdot)$	Big-O notation
$o(\cdot)$	little-o notation
$\square$	end of a proof or solution

Theorem-like Environments

Part I uses theorem-like labels according to the mathematical role of each item. This is especially useful for Science Quiz preparation: a player should be able to tell at a glance whether an item is a definition, a named result, a formula to memorize, or an open problem. Statement-type environments such as Definition, Theorem, Formula, and Algorithm put the heading on its own line, while short guide-type environments such as Idea, Strategy, Tactic, and Remark continue on the same line after the label.

Label	Usage
Definition	a precise meaning of a term or object
Theorem	a major named result, usually worth memorizing by name
Proposition	a useful standard result that is less central than a theorem
Lemma	a supporting result or technical tool
Corollary	a result that follows quickly from a theorem or proposition
Formula	a computational identity or closed form that should be recognized quickly
Algorithm	a step-by-step procedure or named computational method
Fact	a memorization-oriented mathematical or historical fact
Conjecture	a statement believed to be true but not proved in full generality
Hypothesis	a named assumption or guiding claim, often attached to a major open problem
Open Problem	a famous unsolved problem or a problem whose general form remains open
Idea	a short conceptual explanation of why a result is true or useful
Strategy	a fast quiz-solving method or recognition trick
Tactic	a local, immediately usable trick for a particular type of question
Remark	supplementary context, caution, or memorable information

Calculus, Analysis

Calculus notation includes one-variable calculus, multivariable calculus, and vector calculus. The symbol ∇ is read as “nabla” or “del.”

Notation	Meaning
$f'(a)$	the derivative of f at a
$f^{(n)}(a)$	the nth derivative of f at a
$\dfrac{\mathrm{d}y}{\mathrm{d}x}$	the derivative of y with respect to x
$\displaystyle\int_a^b f(x)\,\mathrm{d}x$	the definite integral of f from a to b
$C^k(I)$	the set of k times continuously differentiable functions on I
$C^\infty(I)$	the set of smooth functions on I
∇f or $\operatorname{grad} f$	the gradient of a scalar-valued function f
$\nabla \cdot \mathbf{F}$ or $\operatorname{div} \mathbf{F}$	the divergence of a vector field $\mathbf{F}$
$\nabla \times \mathbf{F}$ or $\operatorname{curl} \mathbf{F}$	the curl of a vector field $\mathbf{F}$
Δf	the Laplacian of f, usually $\nabla \cdot \nabla f$
$D\mathbf{F}$	the derivative matrix, or Jacobian matrix, of a map $\mathbf{F}$
$\operatorname{Jac}(\mathbf{F})$	the Jacobian determinant of $\mathbf{F}$, when defined
$\displaystyle\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$	a line integral of a vector field along a curve C
$\displaystyle\iint_S \mathbf{F} \cdot \mathbf{n}\,\mathrm{d}S$	a flux integral across a surface S

Linear Algebra

In linear algebra, we consistently use bold notation for vectors and matrices. For example, $\mathbf{v} \in \mathbb{R}^n$ is a vector and $\mathbf{A} \in \mathbb{R}^{m\times n}$ is a matrix. The scalar λ is an eigenvalue, not a vector. For linear maps, we use the standard lowercase operator notation $\ker T$ for the kernel and $\operatorname{im} T$ for the image. The notation $\operatorname{Im} z$ is reserved for the imaginary part of a complex number.

Notation	Meaning
$\mathbf{A}^\mathsf{T}$	the transpose of $\mathbf{A}$
$\mathbf{I}_n$	the $n \times n$ identity matrix
$\mathbf{0}$	the zero vector or zero matrix, depending on context
$\det(\mathbf{A})$	the determinant of $\mathbf{A}$
$\operatorname{rk}(\mathbf{A})$	the rank of $\mathbf{A}$
$\operatorname{tr}(\mathbf{A})$	the trace of $\mathbf{A}$
$\operatorname{adj}(\mathbf{A})$	the adjugate of $\mathbf{A}$
$\operatorname{Col}(\mathbf{A})$	the column space of $\mathbf{A}$
$\operatorname{Null}(\mathbf{A})$	the null space of $\mathbf{A}$
$\operatorname{Col}(\mathbf{A}^\mathsf{T})$	the row space of $\mathbf{A}$
$\operatorname{Row}(\mathbf{A})$	another notation for the row space, used only when convenient
$\ker T$	the kernel of a linear map T
$\operatorname{im} T$	the image of a linear map T
$\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$	the span of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$
$\langle \mathbf{u}, \mathbf{v} \rangle$	the inner product of $\mathbf{u}$ and $\mathbf{v}$
$\lVert \mathbf{v} \rVert$	the norm of $\mathbf{v}$
$\mathbf{u} \perp \mathbf{v}$	$\mathbf{u}$ and $\mathbf{v}$ are orthogonal
$U \oplus V$	the direct sum of subspaces U and V

The four fundamental subspaces of $\mathbf{A} \in \mathbb{R}^{m\times n}$ are written as

$$\operatorname{Col}(\mathbf{A}), \qquad \operatorname{Null}(\mathbf{A}), \qquad \operatorname{Col}(\mathbf{A}^\mathsf{T}), \qquad \operatorname{Null}(\mathbf{A}^\mathsf{T}).$$

We use the standard orthogonal decompositions

$$\mathbb{R}^m = \operatorname{Col}(\mathbf{A}) \oplus \operatorname{Null}(\mathbf{A}^\mathsf{T}), \qquad \mathbb{R}^n = \operatorname{Col}(\mathbf{A}^\mathsf{T}) \oplus \operatorname{Null}(\mathbf{A}).$$

For cofactors, we write A_{ij} for the scalar cofactor of the entry a_{ij} of $\mathbf{A}$. The cofactor matrix is denoted by

$$\mathbf{C} = (A_{ij}),$$

and the adjugate matrix is

$$\operatorname{adj}(\mathbf{A}) = \mathbf{C}^\mathsf{T}.$$

We avoid using $\mathbf{A}^*$ for the adjugate, since $\mathbf{A}^*$ often denotes the conjugate transpose in other contexts.

Differential Equations

Differential equations are written using standard derivative notation. The independent variable is usually t or x, depending on the context.

Notation	Meaning
y'	the first derivative of y with respect to the relevant variable
y''	the second derivative of y with respect to the relevant variable
$\dot{x}$	the derivative of x with respect to time t
$\ddot{x}$	the second derivative of x with respect to time t
y_h	the homogeneous part of a solution
y_p	a particular solution
$c_1, c_2, \dots$	arbitrary constants in a general solution
$W(y_1, y_2)(t)$	the Wronskian of y_1 and y_2
$\mathcal{L}\{f(t)\}(s)$	the Laplace transform of $f(t)$

Combinatorics and Discrete Mathematics, Probability and Statistics

Combinatorics and discrete mathematics provide the notation for counting, recurrence relations, and graphs. Probability and statistics use this notation frequently, especially in finite probability spaces.

Notation	Meaning
$n!$	factorial
$\binom{n}{k}$	the number of k-element subsets of an n-element set
$P(n, k)$	the number of k-permutations of an n-element set
a_n	the nth term of a sequence
$G = (V, E)$	a graph with vertex set V and edge set E
$\deg(v)$	the degree of a vertex v in a graph
K_n	the complete graph on n vertices
C_n	the cycle graph on n vertices
P_n	the path graph on n vertices
$\mathbb{P}(A)$	the probability of an event A
$\mathbb{P}(A \mid B)$	the conditional probability of A given B
$\mathbb{E}[X]$	the expectation of a random variable X
$\mathrm{Var}(X)$	the variance of X
$\mathrm{Cov}(X, Y)$	the covariance of X and Y
$X \sim \mathrm{Bin}(n, p)$	X follows a binomial distribution with parameters n and p
$X \sim \mathrm{Poi}(\lambda)$	X follows a Poisson distribution with parameter λ
$X \sim \mathrm{N}(\mu, \sigma^2)$	X follows a normal distribution with mean μ and variance σ^2

Random variables are usually denoted by uppercase letters such as X, Y, Z. Events are denoted by uppercase letters such as A, B, E. In graph-theoretic contexts, $\deg(v)$ denotes the degree of a vertex.

Geometry

We use standard geometric notation unless otherwise stated.

Notation	Meaning
$\triangle ABC$	the triangle with vertices A, B, C
$\angle ABC$	the angle with vertex B
$\overline{AB}$	the line segment joining A and B
AB	the length of the segment $\overline{AB}$, when no confusion is possible
$[ABC]$	the area of triangle ABC, when used in geometry problems
$\parallel$	parallel
$\perp$	perpendicular

Number Theory, Algebra

Unless otherwise stated, divisibility and congruence statements are made in $\mathbb{Z}$. In algebra, the notation is kept standard and lightweight, since Science Quiz rarely requires highly specialized algebraic formalism.

Notation	Meaning
$a \mid b$	a divides b
$a \nmid b$	a does not divide b
$\gcd(a, b)$	the greatest common divisor of a and b
$\operatorname{lcm}(a, b)$	the least common multiple of a and b
$a \equiv b \pmod{n}$	a is congruent to b modulo n
$\varphi(n)$	Euler's phi function, or Euler's totient function
$\operatorname{ord}_n(a)$	the order of a modulo n
p	usually a prime number, when used in number-theoretic contexts
G	usually a group
e	the identity element of a group, when the context is clear
$H \leq G$	H is a subgroup of G
$H \trianglelefteq G$	H is a normal subgroup of G
G/H	the quotient group of G by H
R	usually a ring
F or K	usually a field
$F[x]$	the polynomial ring over a field F
$\deg f$	the degree of a polynomial f
$\operatorname{Aut}(G)$	the automorphism group of G
$\operatorname{Gal}(L/K)$	the Galois group of a field extension L/K

The Chinese Remainder Theorem may be abbreviated as CRT after it is first introduced. In algebraic contexts, $\deg f$ denotes the degree of a polynomial f; in graph-theoretic contexts, $\deg(v)$ denotes the degree of a vertex.

Miscellaneous Topics

Some symbols have different meanings in different areas. In this book, the intended meaning should always be clear from context.

Notation	Meaning
$\lfloor x \rfloor$	the floor of x
$\lceil x \rceil$	the ceiling of x
$\{x\}$	the fractional part of x, usually $x - \lfloor x \rfloor$
RH	the Riemann Hypothesis, after it has been introduced
P vs NP	the P versus NP problem, after it has been introduced

Part I

Mathematical Concepts

Chapter 1

Calculus

Related POSTECH Courses: (MATH101) Calculus I, (MATH102) Calculus II, (MATH210) Applied Complex Variables
Related Books:

One-Variable Calculus

Formula 1.1. Standard Trigonometric and Exponential Limits
The following limits are fundamental:

$$\lim_{x\to 0}\frac{\sin x}{x}=1, \qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2},$$
$$\lim_{x\to 0}\frac{e^x-1}{x}=1, \qquad \lim_{x\to 0}\frac{\log(1+x)}{x}=1,$$

and

$$\lim_{x\to 0}(1+x)^{1/x}=e.$$

Strategy. These limits are often faster than L'Hopital's Rule. In a quiz, the expressions $\sin x$, $1-\cos x$, e^x-1, $\log(1+x)$, and $(1+x)^{1/x}$ are strong signals for these standard patterns.

Theorem 1.2. Intermediate Value Theorem
Let f be continuous on $[a,b]$. If N is any number between $f(a)$ and $f(b)$, then there exists $c\in[a,b]$ such that

$$f(c)=N.$$

Strategy. The Intermediate Value Theorem is the theorem behind many existence arguments. To show that an equation has a solution, define a continuous function and show that its values cross the desired level.

Theorem 1.3. Extreme Value Theorem; Weierstrass Extreme Value Theorem
Let f be continuous on $[a,b]$. Then f attains both a maximum and a minimum on $[a,b]$.

Remark. The closed interval condition is important. A continuous function on an open interval need not attain a maximum or a minimum.

Theorem 1.4. Mean Value Theorem; Lagrange Mean Value Theorem
Let f be continuous on $[a,b]$ and differentiable on (a,b). Then there exists $c\in(a,b)$ such that

$$f'(c)=\frac{f(b)-f(a)}{b-a}.$$

Idea. The Mean Value Theorem says that somewhere on the interval, the instantaneous rate of change equals the average rate of change.

Theorem 1.5. Taylor's Theorem
Suppose that f is $n+1$ times differentiable near a. Then

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x),$$

where, in Lagrange's form of the remainder,

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

for some c between a and x.

Strategy. Taylor expansion is one of the fastest tools for limits, approximations, and hidden series evaluations. Expressions involving e^x, $\sin x$, $\cos x$, $\log(1+x)$, or $\arctan x$ often invite Taylor expansion.

Formula 1.6. Maclaurin Series
The most important Maclaurin series are

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \qquad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!},$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \qquad (|x| < 1),$$

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \qquad (-1 < x \le 1),$$

and

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \qquad (|x| \le 1).$$

Formula 1.7. Integration by Parts
If u and v are differentiable functions, then

$$\int u \, \mathrm{d}v = uv - \int v \, \mathrm{d}u.$$

Equivalently,

$$\int_a^b u(x)v'(x) \, \mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, \mathrm{d}x.$$

Formula 1.8. Arc Length Formula
The length of the graph $y = f(x)$ on $[a, b]$ is

$$L = \int_a^b \sqrt{1 + (f'(x))^2} \, \mathrm{d}x.$$

More generally, if a curve is parametrized by $\mathbf{r}(t) = (x(t), y(t), z(t))$, then

$$L = \int_a^b \|\mathbf{r}'(t)\| \, \mathrm{d}t.$$

Formula 1.9. Viète's Formula
For real $x \neq 0$, we have

$$\frac{\sin x}{x} = \prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right).$$

In particular,

$$\frac{2}{\pi} = \cos\frac{\pi}{4} \cdot \cos\frac{\pi}{8} \cdot \cos\frac{\pi}{16} \cdots .$$

Idea. Viète's formula is a product formula for π. The key identity is the double-angle formula

$$\sin(2x) = 2\sin x \cos x.$$

Formula 1.10. Euler's Formula
For every real number x,

$$e^{ix} = \cos x + i\sin x.$$

In particular,

$$e^{i\pi} + 1 = 0.$$

Remark. The identity $e^{i\pi} + 1 = 0$ is called Euler's identity. It connects the constants e, i, π, 1, and 0 in a single formula.

Formula 1.11. Gaussian Integral
The Gaussian integral is

$$\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}.$$

Equivalently,

$$\int_{0}^{\infty} e^{-x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2}.$$

Remark. The Gaussian integral is one of the most famous integrals in calculus. It is closely connected to the normal distribution in probability.

Formula 1.12. Fresnel Integrals
The Fresnel integrals satisfy

$$\int_{0}^{\infty} \cos(x^2)\,\mathrm{d}x = \int_{0}^{\infty} \sin(x^2)\,\mathrm{d}x = \sqrt{\frac{\pi}{8}}.$$

Remark. Fresnel integrals appear in wave optics, diffraction, and the Cornu spiral. They are memorable examples of non-elementary definite integrals with clean values.

Formula 1.13. Sine Integral; Dirichlet Integral
The sine integral is

$$\mathrm{Si}(x) = \int_{0}^{x} \frac{\sin t}{t}\,\mathrm{d}t.$$

The classical Dirichlet integral is

$$\int_{0}^{\infty} \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}.$$

Formula 1.14. Wallis Product
The Wallis product is

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots .$$

Definition 1.15. Gamma Function
For $x > 0$, the Gamma function is defined by

$$\Gamma(x) = \int_{0}^{\infty} t^{x-1} e^{-t}\,\mathrm{d}t.$$

Proposition 1.16. Gamma Function and Factorials
The Gamma function satisfies

$$\Gamma(x+1) = x\Gamma(x).$$

In particular, for every positive integer n,

$$\Gamma(n) = (n-1)!.$$

Also,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Formula 1.17. Stirling's Formula
As $n \to \infty$,

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

Remark. Stirling's formula is the standard approximation for large factorials. It often appears in counting, probability, and asymptotic estimation.

Formula 1.18. Trapezoidal Rule
The trapezoidal approximation is

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{2}(f(a)+f(b)).$$

It has degree of precision 1.

Formula 1.19. Simpson's Rule
Simpson's rule is

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{6}\left(f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right).$$

It has degree of precision 3.

Remark. Simpson's rule uses quadratic interpolation, but the resulting formula is exact for every polynomial of degree at most 3.

Series Tests

Theorem 1.20. Integral Test
Suppose f is positive, continuous, and decreasing on $[1,\infty)$, and let $a_n = f(n)$. Then

$$\sum_{n=1}^{\infty} a_n$$

and

$$\int_1^{\infty} f(x)\,\mathrm{d}x$$

either both converge or both diverge.

Theorem 1.21. Limit Comparison Test
Let $a_n, b_n > 0$. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

for some finite number $c > 0$, then

$$\sum_{n=1}^{\infty} a_n \qquad \text{and} \qquad \sum_{n=1}^{\infty} b_n$$

converge or diverge together.

Theorem 1.22. Ratio Test; d'Alembert's Ratio Test
Let $a_n > 0$ and suppose

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$$

exists. If $L < 1$, then $\sum a_n$ converges. If $L > 1$, then $\sum a_n$ diverges. If $L = 1$, the test is inconclusive.

Theorem 1.23. Root Test; Cauchy's Root Test
Let $a_n \geq 0$ and suppose

$$L = \lim_{n\to\infty} \sqrt[n]{a_n}$$

exists. If $L < 1$, then $\sum a_n$ converges. If $L > 1$, then $\sum a_n$ diverges. If $L = 1$, the test is inconclusive.

Theorem 1.24. Alternating Series Test; Leibniz Test
If

$$a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$$

and $a_n \to 0$, then the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

converges.

Strategy. For Science Quiz, the test names matter. Factorials and powers often suggest the Ratio Test, nth powers suggest the Root Test, and alternating signs with decreasing terms suggest the Alternating Series Test.

Multivariable Calculus and Vector Calculus

Definition 1.25. Gradient, Divergence, Curl, and Laplacian
For a scalar-valued function $f(x, y, z)$ and a vector field $\mathbf{F} = (P, Q, R)$, we write

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right),$$

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z},$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix},$$

and

$$\Delta f = \nabla \cdot \nabla f = f_{xx} + f_{yy} + f_{zz}.$$

Idea. The gradient points in the direction of steepest increase. Divergence measures source or sink behavior. Curl measures local rotation. The Laplacian measures how a value differs from its surrounding average.

Theorem 1.26. Clairaut's Theorem; Mixed Partial Derivative Theorem
If f has continuous second partial derivatives near a point, then the mixed partial derivatives agree at that point:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Theorem 1.27. Fubini's Theorem
Under suitable continuity or integrability assumptions, an iterated integral over a plane region may be computed in either order. For example, if

$$R = \{(x, y) : a \leq x \leq b,\ g_1(x) \leq y \leq g_2(x)\},$$

then

$$\iint_R f(x, y)\,\mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,\mathrm{d}y\,\mathrm{d}x.$$

Definition 1.28. Jacobian
For a map

$$\mathbf{F}(u_1, \ldots, u_n) = (x_1(u_1, \ldots, u_n), \ldots, x_n(u_1, \ldots, u_n)),$$

the Jacobian matrix is

$$D\mathbf{F} = \left(\frac{\partial x_i}{\partial u_j}\right)_{i,j=1}^{n}.$$

Its determinant

$$\mathrm{Jac}(\mathbf{F}) = \det(D\mathbf{F})$$

is called the Jacobian determinant.

Theorem 1.29. Change of Variables Formula
Under suitable regularity and one-to-one assumptions,

$$\int_{\mathbf{F}(D)} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \int_D f(\mathbf{F}(\mathbf{u}))\,|\det D\mathbf{F}(\mathbf{u})|\,\mathrm{d}\mathbf{u}.$$

Remark. The Jacobian determinant is the local area or volume scaling factor of a change of variables.

Formula 1.30. Polar Coordinates
In polar coordinates,

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad \mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta.$$

The area enclosed by a polar curve $r = r(\theta)$ is

$$A = \frac{1}{2}\int_\alpha^\beta r(\theta)^2\,\mathrm{d}\theta.$$

Theorem 1.31. Lagrange Multipliers
Suppose that f has a local maximum or minimum subject to the constraint

$$g(x_1, \ldots, x_n) = 0$$

at a point where $\nabla g \neq \mathbf{0}$. Then there exists a scalar λ such that

$$\nabla f = \lambda \nabla g.$$

Idea. At a constrained optimum, the level surface of f is tangent to the constraint surface. Thus their normal vectors are parallel.

Definition 1.32. Line Integral
If a curve C is parametrized by $\mathbf{r}(t)$ for $a \leq t \leq b$, then the line integral of a vector field $\mathbf{F}$ along C is

$$\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,\mathrm{d}t.$$

Theorem 1.33. Green's Theorem
Let C be a positively oriented, simple closed curve in the plane, and let D be the region enclosed by C. Then

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}A.$$

Idea. Green's Theorem converts circulation along a boundary curve into curl over the region inside. It is the two-dimensional ancestor of Stokes' Theorem.

Theorem 1.34. Stokes' Theorem
Let S be an oriented smooth surface with positively oriented boundary curve ∂S. Then

$$\oint_{\partial S} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,\mathrm{d}S.$$

Idea. Stokes' Theorem says that circulation around the boundary equals total curl through the surface. It turns a boundary integral into an interior integral.

Theorem 1.35. Divergence Theorem; Gauss's Theorem
Let E be a solid region in $\mathbb{R}^3$ with outward-oriented boundary surface ∂E. Then

$$\iint_{\partial E} \mathbf{F} \cdot \mathbf{n}\,\mathrm{d}S = \iiint_E \nabla \cdot \mathbf{F}\,\mathrm{d}V.$$

Idea. The Divergence Theorem says that outward flux through the boundary equals total source strength inside the region.

Remark. Green's Theorem, Stokes' Theorem, and the Divergence Theorem share the same philosophy: a boundary integral is equal to an interior integral. This is the geometric heart of vector calculus.

Chapter 2

Linear Algebra

Related POSTECH Courses: (MATH203) Applied Linear Algebra
Related Books:

Theorem 2.1. Rank–Nullity Theorem
Let $T : V \to W$ be a linear map between finite-dimensional vector spaces. Then

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

For a matrix $\mathbf{A} \in F^{m\times n}$, this becomes

$$n = \operatorname{rk}(\mathbf{A}) + \dim \operatorname{Null}(\mathbf{A}).$$

Strategy. Rank–Nullity connects variables, pivots, and free variables. It is the linear algebra version of accounting: nothing disappears; dimensions are redistributed.

Theorem 2.2. Row Rank Equals Column Rank
For every matrix $\mathbf{A}$,

$$\dim \operatorname{Row}(\mathbf{A}) = \dim \operatorname{Col}(\mathbf{A}).$$

This common dimension is called the rank of $\mathbf{A}$ and is denoted by $\operatorname{rk}(\mathbf{A})$.

Theorem 2.3. Invertible Matrix Theorem
For a square matrix $\mathbf{A} \in F^{n\times n}$, the following statements are equivalent:

$$\mathbf{A} \text{ is invertible}, \qquad \det(\mathbf{A}) \neq 0, \qquad \operatorname{rk}(\mathbf{A}) = n,$$

$$\operatorname{Null}(\mathbf{A}) = \{\mathbf{0}\},$$

and

$$\mathbf{Ax} = \mathbf{b}$$

has a unique solution for every $\mathbf{b} \in F^n$.

Remark. This theorem is a dictionary. It translates between determinants, rank, null spaces, and solvability of linear systems.

Definition 2.4. Determinant
The determinant is a scalar associated to a square matrix $\mathbf{A} \in F^{n\times n}$, denoted by $\det(\mathbf{A})$. Geometrically, for real matrices, $|\det(\mathbf{A})|$ is the volume scaling factor of the linear transformation $\mathbf{x} \mapsto \mathbf{Ax}$.

Formula 2.5. Determinant as Volume
If $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{R}^n$, then the n-dimensional volume of the parallelepiped spanned by these vectors is

$$\left|\det \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix}\right|.$$

In $\mathbb{R}^3$, this is the absolute value of the scalar triple product

$$|(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}|.$$

Proposition 2.6. Basic Properties of Determinants
For square matrices $\mathbf{A}, \mathbf{B} \in F^{n \times n}$, we have

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}),$$

$$\det(\mathbf{A}^\mathsf{T}) = \det(\mathbf{A}),$$

and, if $\mathbf{A}$ is invertible,

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$

Formula 2.7. Determinant of a Triangular Matrix
If $\mathbf{A}$ is upper triangular or lower triangular with diagonal entries $a_{11}, a_{22}, \dots, a_{nn}$, then

$$\det(\mathbf{A}) = a_{11} a_{22} \cdots a_{nn}.$$

Formula 2.8. Inverse of a 2×2 Matrix
If

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad \text{and} \qquad ad - bc \neq 0,$$

then

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Formula 2.9. Rotation and Reflection Matrices
Rotation by angle θ in the plane is represented by

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

Reflection across the line making angle θ with the positive x-axis is represented by

$$\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}.$$

Proposition 2.10. Trace, Determinant, and Eigenvalues
Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ have eigenvalues $\lambda_1, \dots, \lambda_n$, counted with algebraic multiplicity. Then

$$\operatorname{tr}(\mathbf{A}) = \lambda_1 + \cdots + \lambda_n$$

and

$$\det(\mathbf{A}) = \lambda_1 \lambda_2 \cdots \lambda_n.$$

Definition 2.11. Eigenvalue and Characteristic Polynomial
Let $\mathbf{A} \in F^{n \times n}$. A scalar $\lambda \in F$ is an eigenvalue of $\mathbf{A}$ if there exists a nonzero vector $\mathbf{v} \in F^n$ such that

$$\mathbf{Av} = \lambda \mathbf{v}.$$

The characteristic polynomial of $\mathbf{A}$ is

$$p_{\mathbf{A}}(\lambda) = \det(\lambda \mathbf{I}_n - \mathbf{A}).$$

Proposition 2.12. Eigenvalues and the Characteristic Polynomial
A scalar λ is an eigenvalue of $\mathbf{A}$ if and only if

$$\det(\lambda \mathbf{I}_n - \mathbf{A}) = 0.$$

Equivalently, the eigenvalues of $\mathbf{A}$ are the roots of the characteristic polynomial $p_{\mathbf{A}}(\lambda)$.

Theorem 2.13. Cayley–Hamilton Theorem
Every square matrix satisfies its own characteristic polynomial. That is, if

$$p_{\mathbf{A}}(\lambda) = \det(\lambda \mathbf{I}_n - \mathbf{A}),$$

then

$$p_{\mathbf{A}}(\mathbf{A}) = \mathbf{0}.$$

Remark. Cayley–Hamilton is a famous named theorem. It is especially useful for reducing high powers of a matrix to lower powers.

Theorem 2.14. Diagonalization Criterion
A matrix $\mathbf{A} \in F^{n\times n}$ is diagonalizable over F if and only if F^n has a basis consisting of eigenvectors of $\mathbf{A}$.

Corollary 2.15. Distinct Eigenvalues Imply Diagonalizability
If an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues in F, then $\mathbf{A}$ is diagonalizable over F.

Proposition 2.16. Polynomials of a Diagonalizable Matrix
Let $\mathbf{A} \in F^{n\times n}$ be diagonalizable over F, and suppose

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}, \qquad \mathbf{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

For a polynomial $p(x) \in F[x]$, we have

$$p(\mathbf{A}) = \mathbf{S}p(\mathbf{D})\mathbf{S}^{-1},$$

where

$$p(\mathbf{D}) = \operatorname{diag}\big(p(\lambda_1), \dots, p(\lambda_n)\big).$$

In particular,

$$\det(p(\mathbf{A})) = p(\lambda_1)p(\lambda_2)\cdots p(\lambda_n).$$

Idea. Similar matrices represent the same linear transformation in different bases. Therefore, applying a polynomial to a diagonalizable matrix is the same as applying that polynomial to each diagonal entry after diagonalization.

Strategy. Distinct eigenvalues are an easy sufficient condition for diagonalizability. They are not necessary: a matrix can be diagonalizable even when some eigenvalues are repeated.

Theorem 2.17. Spectral Theorem for Real Symmetric Matrices
If $\mathbf{A} \in \mathbb{R}^{n\times n}$ is symmetric, then all eigenvalues of $\mathbf{A}$ are real, and there exists an orthogonal matrix $\mathbf{Q}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}.$$

Equivalently, $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of $\mathbf{A}$.

Remark. In quiz settings, the words "real symmetric matrix" are a strong signal for real eigenvalues and orthogonal diagonalization.

Theorem 2.18. Spectral Theorem for Hermitian Matrices
If $\mathbf{A} \in \mathbb{C}^{n\times n}$ is Hermitian, meaning

$$\mathbf{A}^* = \mathbf{A},$$

where $\mathbf{A}^*$ is the conjugate transpose, then all eigenvalues of $\mathbf{A}$ are real and there exists a unitary matrix $\mathbf{U}$ such that

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^*$$

for a real diagonal matrix $\mathbf{D}$.

Definition 2.19. Markov Matrix; Stochastic Matrix
A column-stochastic Markov matrix is a matrix whose entries are nonnegative and whose columns each sum to 1. For such a matrix $\mathbf{A}$,

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix} \mathbf{A} = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}.$$

Hence 1 is an eigenvalue of $\mathbf{A}$.

Remark. Markov matrices are linear algebra objects behind finite-state Markov chains. A stationary distribution is an eigenvector corresponding to the eigenvalue 1.

Theorem 2.20. Cauchy–Schwarz Inequality
For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have

$$|\langle \mathbf{u}, \mathbf{v}\rangle| \le \|\mathbf{u}\|\,\|\mathbf{v}\|.$$

Equivalently, for real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \le \left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right).$$

Strategy. The expression $\sum a_i b_i$ is a strong signal for Cauchy–Schwarz. The vector form is often the fastest way to recognize the inequality.

Corollary 2.21. Triangle Inequality
For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have

$$\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Proposition 2.22. Properties of Orthogonal Matrices
A real square matrix $\mathbf{Q}$ is orthogonal if

$$\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}.$$

If $\mathbf{Q}$ is orthogonal, then

$$\|\mathbf{Q}\mathbf{v}\| = \|\mathbf{v}\|, \qquad \langle \mathbf{Q}\mathbf{u}, \mathbf{Q}\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle, \qquad \det(\mathbf{Q}) = \pm 1.$$

Algorithm 2.23. Gaussian Elimination
Gaussian elimination uses elementary row operations to transform a matrix into an upper triangular or row echelon form. It is the standard computational method for solving linear systems, finding ranks, and computing inverses.

Theorem 2.24. LU Decomposition
When Gaussian elimination can be performed without row exchanges, a square matrix $\mathbf{A}$ can be written as

$$\mathbf{A} = \mathbf{L}\mathbf{U},$$

where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.

Theorem 2.25. Gram–Schmidt Process; QR Decomposition
From any linearly independent list

$$\mathbf{v}_1, \ldots, \mathbf{v}_k$$

in an inner product space, one can construct an orthonormal list

$$\mathbf{q}_1, \ldots, \mathbf{q}_k$$

with the same span. In matrix form, this leads to the QR decomposition

$$\mathbf{A} = \mathbf{Q}\mathbf{R},$$

where the columns of $\mathbf{Q}$ *are orthonormal and* $\mathbf{R}$ *is upper triangular.*

Remark. Gram–Schmidt is the standard procedure for turning an arbitrary basis into an orthonormal basis.

Formula 2.26. Projection Matrix
If the columns of $\mathbf{Q} \in \mathbb{R}^{n \times k}$ are orthonormal, then the orthogonal projection matrix onto $\mathrm{Col}(\mathbf{Q})$ is

$$\mathbf{P} = \mathbf{Q}\mathbf{Q}^\mathsf{T}.$$

More generally, if the columns of $\mathbf{A} \in \mathbb{R}^{n \times k}$ are linearly independent, then the orthogonal projection matrix onto $\mathrm{Col}(\mathbf{A})$ is

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}.$$

Proposition 2.27. Projection Matrix Criterion
A matrix $\mathbf{P}$ *is a projection matrix if*

$$\mathbf{P}^2 = \mathbf{P}.$$

It is an orthogonal projection matrix if, in addition,

$$\mathbf{P}^\mathsf{T} = \mathbf{P}.$$

Definition 2.28. Positive Definite Matrix
A real symmetric matrix $\mathbf{A}$ is positive definite if

$$\mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x} > 0$$

for every nonzero vector $\mathbf{x}$.

Theorem 2.29. Sylvester's Criterion
A real symmetric matrix is positive definite if and only if all of its leading principal minors are positive.

Definition 2.30. Singular Values
Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The singular values of $\mathbf{A}$ are the nonnegative square roots of the eigenvalues of

$$\mathbf{A}^\mathsf{T}\mathbf{A}.$$

Theorem 2.31. Singular Value Decomposition; SVD
Every real matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ *can be written as*

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\mathsf{T},$$

where $\mathbf{U}$ *and* $\mathbf{V}$ *are orthogonal matrices and* $\boldsymbol{\Sigma}$ *is a rectangular diagonal matrix whose diagonal entries are the singular values of* $\mathbf{A}$*.*

Remark. Singular Value Decomposition applies to every real matrix, not only to square or diagonalizable matrices. This is why SVD is one of the most useful decompositions in applied linear algebra.

Theorem 2.32. Jordan Canonical Form
Over the complex numbers, every square matrix is similar to a Jordan matrix. That is, for every $\mathbf{A} \in \mathbb{C}^{n\times n}$*, there exists an invertible matrix* $\mathbf{P}$ *such that*

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

is in Jordan canonical form.

Remark. Jordan canonical form is usually more important as a named concept than as a computational tool in Science Quiz. It describes what remains when a matrix cannot be fully diagonalized.

Chapter 3

Differential Equations

Related POSTECH Courses: (MATH200) Differential Equations
Related Books:

Theorem 3.1. Picard–Lindelof Existence and Uniqueness Theorem
Consider the initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

If f and $\partial f/\partial y$ are continuous near (t_0, y_0), then there exists a unique solution near $t = t_0$.

Remark. In a quiz setting, the important phrase is "existence and uniqueness." It tells us when an initial condition determines exactly one solution.

Theorem 3.2. Superposition Principle
If $y_1, \dots, y_k$ are solutions of a homogeneous linear differential equation, then any linear combination

$$c_1y_1 + \cdots + c_ky_k$$

is also a solution.

Remark. The Superposition Principle is the differential-equation version of linear algebra. It is one of the main reasons linear equations are much easier to handle than nonlinear equations.

Proposition 3.3. General Solution of a Nonhomogeneous Linear Equation
For a nonhomogeneous linear differential equation, the general solution has the form

$$y = y_h + y_p,$$

where y_h is the general solution of the associated homogeneous equation and y_p is one particular solution of the nonhomogeneous equation.

Formula 3.4. Second-Order Constant-Coefficient Equation
Consider the homogeneous equation

$$ay'' + by' + cy = 0, \qquad a \neq 0.$$

Its characteristic equation is

$$ar^2 + br + c = 0.$$

If the roots are distinct real numbers r_1, r_2, then

$$y_h = c_1e^{r_1t} + c_2e^{r_2t}.$$

If the equation has a repeated real root r, then

$$y_h = (c_1 + c_2t)e^{rt}.$$

If the roots are $\alpha \pm i\beta$ with $\beta \neq 0$, then

$$y_h = e^{\alpha t}\left(c_1 \cos \beta t + c_2 \sin \beta t\right).$$

Remark. The characteristic equation converts a constant-coefficient linear ODE into an algebraic equation. This is the key trick.

Definition 3.5. Wronskian
For two differentiable functions y_1 and y_2, their Wronskian is

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t).$$

Proposition 3.6. Wronskian and Linear Independence
If

$$W(y_1, y_2)(t_0) \neq 0$$

for some t_0, then y_1 and y_2 are linearly independent.

Formula 3.7. Exponential Growth and Decay
The equation

$$y' = ky$$

has solution

$$y(t) = Ce^{kt}.$$

If $k > 0$, this models exponential growth. If $k < 0$, this models exponential decay.

Formula 3.8. Logistic Equation
The logistic equation is

$$y' = ky\left(1 - \frac{y}{K}\right),$$

where $k > 0$ is the growth rate and $K > 0$ is the carrying capacity. Its equilibrium solutions are

$$y = 0 \qquad \text{and} \qquad y = K.$$

Remark. The logistic equation is the standard model for population growth with limited resources. The parameter K represents the limiting population.

Formula 3.9. Simple Harmonic Oscillator
The equation

$$x'' + \omega^2 x = 0$$

has general solution

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t.$$

Equivalently,

$$x(t) = A\cos(\omega t - \phi)$$

for suitable constants A and ϕ.

Remark. This is the equation behind ideal mass–spring motion and many small oscillation models in physics.

Formula 3.10. Damped Harmonic Oscillator
The equation

$$mx'' + cx' + kx = 0$$

models a damped harmonic oscillator. Its characteristic equation is

$$mr^2 + cr + k = 0.$$

The motion is underdamped, critically damped, or overdamped according as

$$c^2 - 4mk < 0, \qquad c^2 - 4mk = 0, \qquad c^2 - 4mk > 0.$$

Definition 3.11. Laplace Transform
The Laplace transform of a function $f(t)$ is

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty e^{-st} f(t)\,\mathrm{d}t.$$

Proposition 3.12. Basic Laplace Transforms
The most common Laplace transforms are

$$\mathcal{L}\{1\}(s) = \frac{1}{s}, \qquad \mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a},$$
$$\mathcal{L}\{\sin at\}(s) = \frac{a}{s^2+a^2}, \qquad \mathcal{L}\{\cos at\}(s) = \frac{s}{s^2+a^2}.$$

Remark. The Laplace transform turns many differential equations into algebraic equations. This is its main computational advantage.

Definition 3.13. Heat Equation
The heat equation is the partial differential equation

$$u_t = \alpha u_{xx}, \qquad \alpha > 0.$$

It models diffusion, especially the flow of heat in a one-dimensional medium.

Definition 3.14. Wave Equation
The one-dimensional wave equation is

$$u_{tt} = c^2 u_{xx}.$$

It models waves traveling with speed c.

Theorem 3.15. d'Alembert's Formula
The solution of the wave equation

$$u_{tt} = c^2 u_{xx}$$

on the real line with initial conditions

$$u(x,0) = f(x), \qquad u_t(x,0) = g(x),$$

is

$$u(x,t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,\mathrm{d}s.$$

Remark. d'Alembert's formula shows that waves move along the characteristic lines $x - ct =$ constant and $x + ct =$ constant.

Definition 3.16. Laplace Equation
Laplace's equation is

$$\Delta u = 0.$$

In two variables, this means

$$u_{xx} + u_{yy} = 0.$$

A function satisfying Laplace's equation is called harmonic.

Definition 3.17. Poisson Equation
Poisson's equation is

$$\Delta u = f.$$

It is an inhomogeneous version of Laplace's equation.

Remark. The heat equation, wave equation, and Laplace equation are three of the most famous PDEs. A quiz problem may ask for their names, forms, or physical interpretations.

Chapter 4

Combinatorics and Discrete Mathematics

Related POSTECH Courses: (MATH261) Discrete Mathematics
Related Books:

Formula 4.1. Binomial Theorem; Newton's Binomial Formula
For every nonnegative integer n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

In particular,

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

Remark. The coefficients $\binom{n}{k}$ are the entries of Pascal's triangle. This is one of the most recognizable bridges between algebra and counting.

Theorem 4.2. Pigeonhole Principle; Dirichlet's Box Principle
If $n+1$ objects are placed into n boxes, then at least one box contains at least two objects. More generally, if N objects are placed into k boxes, then at least one box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

Strategy. The Pigeonhole Principle is often hidden in problems asking us to prove that two objects must share a property. The hard part is usually choosing the right "boxes."

Theorem 4.3. Inclusion–Exclusion Principle; Sieve Formula
For finite sets $A_1, \dots, A_n$,

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|.$$

Remark. Inclusion–Exclusion is the standard correction mechanism for overcounting. It is also historically connected to sieve methods.

Formula 4.4. Derangement Formula
Let D_n be the number of derangements of n objects, that is, permutations with no fixed points. Then

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Moreover,

$$D_n = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor.$$

Remark. The approximation $D_n \approx n!/e$ is very memorable. In probability language, a random permutation has no fixed point with probability approximately $1/e$.

Formula 4.5. Catalan Numbers
The nth Catalan number is

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

It begins as

$$1,\ 1,\ 2,\ 5,\ 14,\ 42,\ 132, \ldots.$$

Remark. Catalan numbers count many different objects, including correct parenthesizations, Dyck paths, triangulations of a polygon, and rooted full binary trees. The number 42 is a classic giveaway.

Theorem 4.6. Cayley's Formula
The number of labeled trees on n vertices is

$$n^{n-2}.$$

Remark. Cayley's formula is a perfect Science Quiz result: the statement is short, the answer is surprising, and the name is famous.

Theorem 4.7. Burnside's Lemma; Cauchy–Frobenius Lemma
Suppose a finite group G acts on a finite set X. The number of orbits is

$$\frac{1}{|G|}\sum_{g\in G} |\operatorname{Fix}(g)|,$$

where

$$\operatorname{Fix}(g) = \{x \in X : g \cdot x = x\}.$$

Idea. Burnside's Lemma says that the number of essentially different objects is the average number of objects fixed by a symmetry.

Lemma 4.8. Handshaking Lemma
For a finite undirected graph $G = (V, E)$,

$$\sum_{v\in V} \deg(v) = 2|E|.$$

In particular, the number of vertices of odd degree is even.

Remark. The name comes from the idea that every handshake contributes 2 to the total count of hands shaken.

Theorem 4.9. Euler's Formula for Planar Graphs
If a connected planar graph has V vertices, E edges, and F faces, then

$$V - E + F = 2.$$

Remark. This is the graph-theoretic version of Euler's polyhedron formula. It is one of the most famous formulas involving planar graphs.

Theorem 4.10. Eulerian Trail Criterion
A connected finite graph has an Eulerian circuit if and only if every vertex has even degree. It has an Eulerian trail but not an Eulerian circuit if and only if exactly two vertices have odd degree.

Remark. This theorem is historically connected to Euler's solution of the Seven Bridges of Konigsberg problem.

Theorem 4.11. Hall's Marriage Theorem
Let $G = (X \cup Y, E)$ be a bipartite graph. There is a matching that covers every vertex of X if and only if, for every subset $S \subseteq X$,

$$|N(S)| \geq |S|,$$

where $N(S)$ is the set of neighbors of vertices in S.

Strategy. Hall's Marriage Theorem is the standard named theorem for proving that a perfect assignment or matching exists.

Theorem 4.12. Ramsey's Theorem
For any positive integers r and s, there exists an integer $R(r, s)$ such that every red-blue coloring of the edges of a sufficiently large complete graph contains either a red K_r or a blue K_s.

Formula 4.13. The Party Problem
The classical Ramsey number

$$R(3, 3) = 6$$

says that among any six people, there are either three mutual acquaintances or three mutual strangers.

Remark. Ramsey theory is the mathematics of unavoidable structure. Its slogan is that complete disorder is impossible in a large enough system.

Theorem 4.14. Erdos–Szekeres Theorem
Any sequence of

$$(r-1)(s-1)+1$$

distinct real numbers contains either an increasing subsequence of length r or a decreasing subsequence of length s.

Remark. This is a beautiful pigeonhole-style result. It is often remembered as the monotone subsequence theorem.

Chapter 5

Probability and Statistics

Related POSTECH Courses: (MATH230) Probability & Statistics
Related Books:

Theorem 5.1. Law of Total Probability
If $B_1, \dots, B_n$ form a partition of the sample space and $\mathbb{P}(B_i) > 0$ for all i, then

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Theorem 5.2. Bayes' Theorem
If $B_1, \dots, B_n$ form a partition of the sample space and $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

In the two-event form,

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Remark. Bayes' Theorem is the standard formula for reversing conditional probability. It updates a prior probability after observing evidence.

Proposition 5.3. Linearity of Expectation
For random variables $X_1, \dots, X_n$,

$$\mathbb{E}[X_1 + \cdots + X_n] = \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n].$$

This holds even when the random variables are not independent.

Strategy. Linearity of expectation is often the fastest way to count an expected number of objects. Introduce indicator random variables and add their expectations.

Formula 5.4. Variance and Covariance Identities
For a random variable X,

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

For random variables X and Y,

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y).$$

If X and Y are independent, then

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y).$$

Formula 5.5. Binomial Distribution
If $X \sim \mathrm{Bin}(n, p)$, then

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Its mean and variance are

$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p).$$

Formula 5.6. Poisson Distribution
If $X \sim \mathrm{Poi}(\lambda)$, then

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots.$$

Its mean and variance are

$$\mathbb{E}[X] = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

Theorem 5.7. Poisson Limit Theorem; Law of Small Numbers
If $n \to \infty$, $p \to 0$, and $np \to \lambda$, then

$$\binom{n}{k} p^k (1-p)^{n-k} \to e^{-\lambda} \frac{\lambda^k}{k!}$$

for each fixed $k \geq 0$.

Remark. This theorem explains why the Poisson distribution models rare events: many trials, small success probability, and finite expected count.

Formula 5.8. Normal Distribution; Gaussian Distribution
A random variable $X \sim \mathrm{N}(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The standard normal distribution is $\mathrm{N}(0, 1)$.

Formula 5.9. Empirical Rule; 68–95–99.7 Rule
For a normal distribution, approximately

$$68\%$$

of the mass lies within one standard deviation of the mean,

$$95\%$$

lies within two standard deviations, and

$$99.7\%$$

lies within three standard deviations.

Theorem 5.10. Weak Law of Large Numbers
Let $X_1, X_2, \ldots$ be independent identically distributed random variables with mean μ. Then, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{X_1 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right) \to 0$$

as $n \to \infty$.

Idea. The Law of Large Numbers says that the sample average stabilizes near the true mean when the number of trials becomes large.

Theorem 5.11. Central Limit Theorem
Let $X_1, X_2, \dots$ be independent identically distributed random variables with mean μ and variance $\sigma^2 > 0$. Then

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution to the standard normal distribution $\mathrm{N}(0,1)$.

Remark. The Central Limit Theorem explains why the normal distribution appears so often in nature and statistics.

Theorem 5.12. Markov's Inequality
If X is a nonnegative random variable and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Theorem 5.13. Chebyshev's Inequality
If X has mean μ and variance σ^2, then for every $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

Remark. Markov's inequality uses only the mean, while Chebyshev's inequality uses the variance. Both are probability bounds that work without assuming a normal distribution.

Theorem 5.14. Jensen's Inequality
If φ is convex, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

If φ is concave, the inequality is reversed.

Idea. Jensen's inequality says that, for a convex function, applying the function after averaging gives a value no larger than averaging after applying the function.

Formula 5.15. Birthday Problem Approximation
In a group of n people, assuming 365 equally likely birthdays, the probability that at least two people share a birthday is approximately

$$1 - \exp\left(-\frac{n(n-1)}{730}\right).$$

For $n = 23$, this probability is already greater than $1/2$.

Remark. The birthday problem is famous because the answer is surprisingly small: only 23 people are needed for a better-than-even chance of a shared birthday.

Formula 5.16. Gambler's Ruin
Suppose a gambler starts with i dollars and stops when reaching either 0 dollars or N dollars. If the probability of winning each round is p and $q = 1 - p$, then the probability of reaching N before 0 is

$$\frac{1 - (q/p)^i}{1 - (q/p)^N} \qquad (p \neq q).$$

In the fair case $p = q = 1/2$, the probability is

$$\frac{i}{N}.$$

Remark. Gambler's ruin is a classic random walk result. It is a good example where the fair case has a very simple answer.

Chapter 6

Geometry

Related POSTECH Courses:
Related Books:

Synthetic Geometry

Definition 6.1. Classical Triangle Centers
For a triangle $\triangle ABC$, the most important classical centers are:

$$O = \text{circumcenter}, \qquad I = \text{incenter}, \qquad H = \text{orthocenter}, \qquad G = \text{centroid}.$$

Here O is the center of the circumcircle, I is the center of the incircle, H is the intersection of the altitudes, and G is the intersection of the medians.

Remark. Triangle centers are extremely common in olympiad-style geometry. The symbols O, I, H, G are standard and should be recognized quickly.

Formula 6.2. Extended Law of Sines
For a triangle $\triangle ABC$ with side lengths

$$a = BC, \qquad b = CA, \qquad c = AB,$$

and circumradius R, we have

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Formula 6.3. Heron's Formula
Let $\triangle ABC$ have side lengths a, b, c and semiperimeter

$$s = \frac{a+b+c}{2}.$$

Then its area K is

$$K = \sqrt{s(s-a)(s-b)(s-c)}.$$

Also,

$$K = rs \qquad \text{and} \qquad K = \frac{abc}{4R},$$

where r is the inradius and R is the circumradius.

Strategy. The formulas

$$K = rs, \qquad K = \frac{abc}{4R}$$

are often more useful than Heron's formula when incircles or circumcircles appear.

Theorem 6.4. Angle Bisector Theorem
In a triangle $\triangle ABC$, suppose the internal angle bisector of $\angle A$ meets BC at D. Then

$$\frac{BD}{DC} = \frac{AB}{AC}.$$

Theorem 6.5. Ceva's Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively. Then the cevians AD, BE, CF are concurrent if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1,$$

using directed lengths.

Remark. Ceva's Theorem is the standard theorem for proving concurrence of three lines in a triangle.

Theorem 6.6. Menelaus' Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively. Then the points D, E, F are collinear if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1,$$

using directed lengths.

Remark. Ceva is for concurrence, while Menelaus is for collinearity. This pair is one of the most important toolkits in olympiad geometry.

Theorem 6.7. Euler Line Theorem
In any non-equilateral triangle, the circumcenter O, centroid G, and orthocenter H are collinear. Moreover,

$$OG : GH = 1 : 2.$$

The line passing through these points is called the Euler line.

Theorem 6.8. Nine-Point Circle Theorem; Euler Circle
In any triangle, the following nine points lie on one circle: the three side midpoints, the three feet of the altitudes, and the three midpoints between the orthocenter and the vertices. This circle is called the nine-point circle.

Remark. The center of the nine-point circle is the midpoint of the segment joining the circumcenter O and the orthocenter H.

Theorem 6.9. Euler's Formula for the Incenter and Circumcenter
For a triangle with circumradius R, inradius r, circumcenter O, and incenter I, we have

$$OI^2 = R^2 - 2Rr.$$

In particular,

$$R \geq 2r.$$

Remark. The inequality $R \geq 2r$ is called Euler's inequality for triangles. Equality holds exactly for an equilateral triangle.

Theorem 6.10. Power of a Point Theorem
Let P be a point and let a line through P meet a circle at A and B. The product

$$PA \cdot PB$$

is independent of the choice of the line through P. It is called the power of P with respect to the circle.

Formula 6.11. Tangent–Secant Theorem
If PT is tangent to a circle at T and a secant through P meets the circle at A and B, then

$$PT^2 = PA \cdot PB.$$

Theorem 6.12. Radical Axis Theorem
For two nonconcentric circles, the set of points having equal powers with respect to the two circles is a line. This line is called the radical axis of the two circles.

Remark. Radical axes are powerful because they turn circle configurations into collinearity statements.

Theorem 6.13. Ptolemy's Theorem
A quadrilateral $ABCD$ is cyclic if and only if

$$AC \cdot BD = AB \cdot CD + BC \cdot DA$$

for the appropriate cyclic order of the vertices.

Remark. Ptolemy's Theorem is one of the most recognizable formulas for cyclic quadrilaterals.

Theorem 6.14. Simson Line Theorem
Let P be a point on the circumcircle of $\triangle ABC$. The feet of the perpendiculars from P to the three lines AB, BC, CA are collinear. This line is called the Simson line of P with respect to $\triangle ABC$.

Theorem 6.15. Miquel's Theorem
In a complete quadrilateral, the circumcircles of the four triangles formed by choosing three of the four lines pass through a common point. This point is called the Miquel point.

Remark. Miquel's Theorem is a standard source of hidden cyclic quadrilaterals in olympiad geometry.

Theorem 6.16. Butterfly Theorem
Let M be the midpoint of a chord PQ of a circle. Through M, draw two other chords AB and CD, and let AD and BC meet PQ at X and Y, respectively. Then M is also the midpoint of XY.

Theorem 6.17. Morley's Trisector Theorem
In any triangle, the adjacent trisectors of the three angles intersect in three points that form an equilateral triangle.

Remark. Morley's theorem is famous because the conclusion is surprisingly simple: angle trisectors always create an equilateral triangle.

Theorem 6.18. Napoleon's Theorem
Construct equilateral triangles externally on the three sides of any triangle. Then the centers of these three equilateral triangles form an equilateral triangle.

Theorem 6.19. Classification of Platonic Solids
There are exactly five Platonic solids:

tetrahedron, cube, octahedron, dodecahedron, icosahedron.

Remark. The classification of Platonic solids is a classic result in Euclidean geometry and a very natural quiz fact.

Analytic Geometry

Formula 6.20. Shoelace Formula; Gauss's Area Formula
For a polygon with vertices

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

in cyclic order, its area is

$$\frac{1}{2}\left|\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1} y_i)\right|,$$

where

$$(x_{n+1}, y_{n+1}) = (x_1, y_1).$$

Remark. The Shoelace Formula is one of the cleanest examples of analytic geometry: a geometric area becomes a determinant-like coordinate calculation.

Definition 6.21. Conic Section
A conic section is a curve obtained by intersecting a plane with a double cone. The nondegenerate conics are:

ellipse, parabola, hyperbola.

Equivalently, a conic can be described using a focus, a directrix, and an eccentricity e.

Formula 6.22. Focus–Directrix Definition of Conics
A conic is the set of points P satisfying

$$\frac{\operatorname{dist}(P, \text{focus})}{\operatorname{dist}(P, \text{directrix})} = e.$$

It is an ellipse if

$$0 < e < 1,$$

a parabola if

$$e = 1,$$

and a hyperbola if

$$e > 1.$$

Formula 6.23. Standard Forms of Conics
The standard form of an ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

The standard form of a hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

The standard form of a parabola opening to the right is

$$y^2 = 4px.$$

Formula 6.24. Eccentricity of an Ellipse and a Hyperbola
For an ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad (a \geq b > 0),$$

the eccentricity is

$$e = \frac{c}{a}, \qquad c^2 = a^2 - b^2.$$

For a hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

the eccentricity is

$$e = \frac{c}{a}, \qquad c^2 = a^2 + b^2.$$

Theorem 6.25. Reflection Property of Conics
An ellipse reflects a ray from one focus to the other focus. A parabola reflects rays parallel to its axis through its focus. A hyperbola has the analogous reflection property involving its two foci and branches.

Remark. The reflection property gives a geometric reason why conics appear in optics, satellite dishes, whispering galleries, and orbital mechanics.

Theorem 6.26. Quadratic Classification of Conics
Consider a real quadratic equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

The discriminant

$$B^2 - 4AC$$

determines the type of the nondegenerate conic:

$$B^2 - 4AC < 0 \quad \Rightarrow \quad \textit{ellipse-type},$$

$$B^2 - 4AC = 0 \quad \Rightarrow \quad \textit{parabola-type},$$

and

$$B^2 - 4AC > 0 \quad \Rightarrow \quad \textit{hyperbola-type}.$$

Remark. This is the analytic-geometry counterpart of classifying conic sections by their shapes.

Formula 6.27. Polar Equation of a Conic
With a focus at the pole, a conic has polar equation

$$r = \frac{\ell}{1 + e\cos\theta}$$

after choosing a suitable axis. Here e is the eccentricity and ℓ is the semi-latus rectum.

Remark. The same equation describes ellipses, parabolas, and hyperbolas depending on whether $e < 1$, $e = 1$, or $e > 1$.

Definition 6.28. Apollonius Circle
Given two fixed points A and B and a positive constant $k \neq 1$, the locus of points P satisfying

$$\frac{PA}{PB} = k$$

is a circle. This circle is called an Apollonius circle.

Remark. Apollonius circles turn a ratio of distances into a concrete circle, which is a common hidden structure in geometry problems.

Theorem 6.29. Descartes' Circle Theorem; Soddy Circle Theorem
Suppose four circles are mutually tangent, and let their curvatures be

$$k_1, k_2, k_3, k_4,$$

where curvature means the reciprocal of the radius, with sign convention for an enclosing circle. Then

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2).$$

Remark. Descartes' Circle Theorem is a striking formula: tangency of four circles becomes one quadratic equation in their curvatures.

Theorem 6.30. Pick's Theorem
Let P be a simple polygon whose vertices lie on lattice points in the plane. If I is the number of lattice points strictly inside P and B is the number of lattice points on the boundary of P, then

$$\text{Area}(P) = I + \frac{B}{2} - 1.$$

Remark. Pick's Theorem is a beautiful bridge between geometry and arithmetic: area is determined by counting lattice points.

Chapter 7

Number Theory

Related POSTECH Courses: (MATH304) Introduction to Number Theory
Related Books:

Theorem 7.1. Fundamental Theorem of Arithmetic
Every integer $n > 1$ can be written as a product of prime numbers. Moreover, this factorization is unique up to the order of the factors. Equivalently,

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where $p_1, \dots, p_k$ are distinct primes and $\alpha_1, \dots, \alpha_k$ are positive integers, and this expression is unique up to reordering.

Remark. This theorem is often called unique prime factorization. It is the reason prime numbers are treated as the basic building blocks of integers.

Theorem 7.2. Euclid's Theorem; Infinitude of Primes
There are infinitely many prime numbers.

Idea. If $p_1, \dots, p_n$ are finitely many primes, consider

$$p_1 p_2 \cdots p_n + 1.$$

This number is not divisible by any prime on the list.

Theorem 7.3. Bezout's Identity
Let $a, b \in \mathbb{Z}$, not both zero. Then there exist integers $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b).$$

In particular, if $\gcd(a, b) = 1$, then there exist integers $x, y \in \mathbb{Z}$ such that

$$ax + by = 1.$$

Remark. Bezout's identity is the algebraic heart of the Euclidean algorithm and modular inverses.

Algorithm 7.4. Euclidean Algorithm
To compute $\gcd(a, b)$ for integers $a, b \in \mathbb{Z}$, repeatedly divide the larger number by the smaller number and replace the pair by

$$(b, r),$$

where r is the remainder. Continue until the remainder becomes 0. The last nonzero remainder is $\gcd(a, b)$.

Theorem 7.5. Chinese Remainder Theorem; CRT
Let $m_1, \ldots, m_k$ be pairwise relatively prime positive integers. Then the system

$$x \equiv a_1 \pmod{m_1}, \qquad x \equiv a_2 \pmod{m_2}, \qquad \ldots, \qquad x \equiv a_k \pmod{m_k}$$

has a solution. Moreover, the solution is unique modulo

$$M = m_1 m_2 \cdots m_k.$$

Strategy. CRT is the standard theorem for combining several congruence conditions into one congruence.

Formula 7.6. Euler's Phi Function Formula
If

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

is the prime factorization of n, then

$$\varphi(n) = n \prod_{i=1}^{k} \left(1 - \frac{1}{p_i}\right).$$

In particular, for a prime power,

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p-1).$$

Theorem 7.7. Fermat's Little Theorem
Let p be a prime number. If $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for every integer a,

$$a^p \equiv a \pmod{p}.$$

Theorem 7.8. Euler's Theorem
If $\gcd(a, n) = 1$, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Remark. Fermat's Little Theorem is the special case of Euler's Theorem when $n = p$ is prime, since $\varphi(p) = p - 1$.

Theorem 7.9. Wilson's Theorem
A positive integer $p > 1$ is prime if and only if

$$(p-1)! \equiv -1 \pmod{p}.$$

Remark. Wilson's theorem is famous because it gives a clean factorial characterization of primes, even though it is not efficient as a primality test.

Definition 7.10. Legendre Symbol
Let p be an odd prime. The Legendre symbol is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1, & \text{if } a \text{ is a quadratic nonresidue modulo } p, \\ 0, & \text{if } p \mid a. \end{cases}$$

Theorem 7.11. Euler's Criterion
Let p be an odd prime. For any integer a,

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

Equivalently, if $p \nmid a$, then

$$a^{(p-1)/2} \equiv \begin{cases} 1 \pmod{p}, & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 \pmod{p}, & \text{if } a \text{ is a quadratic nonresidue modulo } p. \end{cases}$$

Theorem 7.12. Law of Quadratic Reciprocity; Gauss's Theorem of Quadratic Reciprocity
Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

Equivalently,

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$$

unless both p and q are congruent to 3 modulo 4.

Remark. Quadratic reciprocity is one of the crown jewels of elementary number theory. Gauss called it the golden theorem.

Theorem 7.13. Fermat's Two-Square Theorem
An odd prime p can be written as a sum of two squares,

$$p = a^2 + b^2$$

for some integers $a, b \in \mathbb{Z}$, if and only if

$$p \equiv 1 \pmod{4}.$$

Also,

$$2 = 1^2 + 1^2.$$

Theorem 7.14. Lagrange's Four-Square Theorem
Every nonnegative integer can be written as a sum of four integer squares. That is, for every $n \in \mathbb{N}_0$, there exist integers $a, b, c, d \in \mathbb{Z}$ such that

$$n = a^2 + b^2 + c^2 + d^2.$$

Remark. The two-square theorem has a congruence condition. The four-square theorem has no exception at all.

Theorem 7.15. Dirichlet's Theorem on Arithmetic Progressions
If a and d are relatively prime positive integers, then the arithmetic progression

$$a,\ a+d,\ a+2d,\ a+3d, \ldots$$

contains infinitely many primes.

Remark. Dirichlet's theorem is a major result connecting primes and arithmetic progressions. The condition $\gcd(a, d) = 1$ is necessary.

Theorem 7.16. Bertrand's Postulate; Chebyshev's Theorem
For every integer $n > 1$, there exists a prime p such that

$$n < p < 2n.$$

Theorem 7.17. Prime Number Theorem
Let $\pi(x)$ be the number of primes less than or equal to x. Then

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{as } x \to \infty.$$

Idea. The Prime Number Theorem says that the "probability" that a large integer near x is prime is roughly $1/\log x$. This is a mental model, not a literal probability statement.

Definition 7.18. Riemann Zeta Function
For $\mathrm{Re}(s) > 1$, the Riemann zeta function is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

It also has the Euler product

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

Remark. The Euler product is the bridge between the zeta function and prime numbers. The Riemann Hypothesis asserts that the nontrivial zeros of $\zeta(s)$ all have real part $1/2$.

Theorem 7.19. Euclid–Euler Theorem on Perfect Numbers
An even positive integer N is perfect if and only if

$$N = 2^{p-1}(2^p - 1),$$

where $2^p - 1$ is a Mersenne prime.

Remark. A perfect number is a positive integer equal to the sum of its positive proper divisors. For example,

$$6 = 1 + 2 + 3.$$

Theorem 7.20. Fermat's Last Theorem
For every integer $n \geq 3$, there do not exist positive integers a, b, c such that

$$a^n + b^n = c^n.$$

Remark. Fermat's Last Theorem was proved by Andrew Wiles. It is one of the most famous theorems in the history of mathematics.

Chapter 8

Algebra

Related POSTECH Courses: (MATH301) Modern Algebra I, (MATH302) Modern Algebra II
Related Books:

Definition 8.1. Group
A group is a set G together with a binary operation satisfying closure, associativity, the existence of an identity element, and the existence of inverses. If the operation is commutative, then G is called an abelian group.

Definition 8.2. Cyclic Group
A group G is cyclic if there exists an element $g \in G$ such that every element of G is a power of g. In this case, we write

$$G = \langle g \rangle.$$

Theorem 8.3. Lagrange's Theorem
If G is a finite group and $H \le G$, then

$$|H| \mid |G|.$$

Equivalently, the order of every subgroup divides the order of the group.

Corollary 8.4. Order of an Element
If G is a finite group and $g \in G$, then the order of g divides $|G|$. In particular,

$$g^{|G|} = e.$$

Theorem 8.5. Cauchy's Theorem
If G is a finite group and a prime p divides $|G|$, then G contains an element of order p.

Theorem 8.6. First Isomorphism Theorem for Groups
If $\varphi : G \to H$ is a group homomorphism, then

$$G/\ker\varphi \cong \operatorname{im}\varphi.$$

Theorem 8.7. Cayley's Theorem
Every group is isomorphic to a subgroup of a symmetric group. In particular, every finite group of order n is isomorphic to a subgroup of S_n.

Theorem 8.8. Sylow Theorems
Let G be a finite group and suppose

$$|G| = p^k m, \qquad p \nmid m,$$

where p is prime. Then G has a subgroup of order p^k, called a Sylow p-subgroup. Moreover, all Sylow p-subgroups are conjugate, and the number n_p of Sylow p-subgroups satisfies

$$n_p \equiv 1 \pmod p, \qquad n_p \mid m.$$

Strategy. For Science Quiz, the most common finite-group quick check is divisibility: subgroup orders and element orders must divide the group order. When a prime divides $|G|$, Cauchy's theorem guarantees an element of that prime order.

Definition 8.9. Ring and Field
A ring is a set equipped with addition and multiplication satisfying the usual distributive laws. A field is a commutative ring in which every nonzero element has a multiplicative inverse.

Definition 8.10. Integral Domain
An integral domain is a commutative ring with identity in which there are no zero divisors. In other words, if $ab = 0$, then $a = 0$ or $b = 0$.

Algorithm 8.11. Polynomial Division Algorithm
Let F be a field. If $f(x), g(x) \in F[x]$ and $g(x) \neq 0$, then there exist unique polynomials $q(x), r(x) \in F[x]$ such that

$$f(x) = q(x)g(x) + r(x), \qquad r(x) = 0 \quad \text{or} \quad \deg r < \deg g.$$

Theorem 8.12. Factor Theorem
For a polynomial $f(x) \in F[x]$ and an element $a \in F$,

$$f(a) = 0 \quad \Longleftrightarrow \quad x - a \textit{ divides } f(x).$$

Theorem 8.13. Fundamental Theorem of Algebra
Every nonconstant complex polynomial has at least one complex root. Equivalently, every polynomial of degree $n \geq 1$ over $\mathbb{C}$ splits into n linear factors over $\mathbb{C}$, counted with multiplicity.

Definition 8.14. Algebraically Closed Field
A field F is algebraically closed if every nonconstant polynomial in $F[x]$ has a root in F.

Theorem 8.15. Galois Correspondence
For a finite Galois extension L/K, there is an inclusion-reversing correspondence between intermediate fields $K \subseteq E \subseteq L$ and subgroups $H \leq \mathrm{Gal}(L/K)$, given by

$$E \longmapsto \mathrm{Gal}(L/E), \qquad H \longmapsto L^H.$$

Theorem 8.16. Abel–Ruffini Theorem
There is no formula for the roots of a general polynomial of degree 5 or higher using only the field operations and radicals.

Remark. The Abel–Ruffini theorem does not say that every quintic equation is unsolvable. It says that no universal radical formula exists for the general quintic.

Chapter 9

Analysis

Related POSTECH Courses: (MATH311) Analysis I, (MATH312) Analysis II
Related Books:

Definition 9.1. Limit Superior and Limit Inferior
For a real sequence (a_n), define

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup_{k\geq n} a_k, \qquad \liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf_{k\geq n} a_k,$$

whenever these extended real limits are allowed.

Theorem 9.2. Monotone Convergence Theorem for Sequences
Every bounded monotone real sequence converges. More precisely, every increasing sequence bounded above converges to its supremum, and every decreasing sequence bounded below converges to its infimum.

Theorem 9.3. Bolzano–Weierstrass Theorem
Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Theorem 9.4. Heine–Borel Theorem
A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Theorem 9.5. Uniform Continuity on Compact Sets
If $K \subseteq \mathbb{R}^n$ is compact and $f : K \to \mathbb{R}^m$ is continuous, then f is uniformly continuous on K.

Theorem 9.6. Weierstrass Extreme Value Theorem
If $K \subseteq \mathbb{R}^n$ is compact and $f : K \to \mathbb{R}$ is continuous, then f attains both a maximum and a minimum on K.

Strategy. In analysis questions, compactness usually means that one may extract a convergent subsequence or that a continuous function attains extrema. In $\mathbb{R}^n$, the fast recognition phrase is "closed and bounded."

Theorem 9.7. Cauchy Criterion
A sequence (a_n) in $\mathbb{R}$ or $\mathbb{C}$ converges if and only if it is Cauchy: for every $\varepsilon > 0$, there exists N such that

$$m, n \geq N \implies |a_m - a_n| < \varepsilon.$$

Theorem 9.8. Weierstrass M-Test
Let (f_n) be a sequence of functions on a set E. If

$$|f_n(x)| \leq M_n \quad \textit{for all } x \in E$$

and

$$\sum_{n=1}^{\infty} M_n$$

converges, then the series

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly on E.

Theorem 9.9. Uniform Limit Theorem
If (f_n) is a sequence of continuous functions on a set E and $f_n \to f$ uniformly on E, then f is continuous on E.

Theorem 9.10. Term-by-Term Differentiation
Suppose f_n are continuously differentiable on an interval and f_n converges at one point. If f_n' converges uniformly, then f_n converges uniformly to a differentiable function f, and

$$f' = \lim_{n\to\infty} f_n'.$$

Theorem 9.11. Arzelà–Ascoli Theorem
A uniformly bounded and equicontinuous family of real-valued continuous functions on a compact metric space has a uniformly convergent subsequence.

Theorem 9.12. Banach Fixed-Point Theorem; Contraction Mapping Theorem
Let (X, d) be a complete metric space, and let $T : X \to X$ be a contraction. Then T has a unique fixed point $x^* \in X$, and the iteration $x_{n+1} = T(x_n)$ converges to x^* for every initial point $x_0 \in X$.

Theorem 9.13. Baire Category Theorem
A complete metric space cannot be written as a countable union of nowhere dense closed sets. Equivalently, a countable intersection of open dense subsets of a complete metric space is dense.

Theorem 9.14. Lebesgue Dominated Convergence Theorem
If $f_n \to f$ almost everywhere and there exists an integrable function g such that $|f_n| \le g$ for all n, then

$$\lim_{n\to\infty} \int f_n \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu.$$

Theorem 9.15. Fubini's Theorem
Under suitable integrability assumptions, an iterated integral may be computed in either order:

$$\int_X \left(\int_Y f(x, y) \, \mathrm{d}y \right) \mathrm{d}x = \int_Y \left(\int_X f(x, y) \, \mathrm{d}x \right) \mathrm{d}y.$$

Chapter 10

Topology

Related POSTECH Courses: (MATH321) General Topology
Related Books:

Definition 10.1. Topology
A topology on a set X is a collection $\mathcal{T}$ of subsets of X whose elements are called open sets, satisfying:

(i) $\varnothing, X \in \mathcal{T}$;

(ii) arbitrary unions of sets in $\mathcal{T}$ belong to $\mathcal{T}$;

(iii) finite intersections of sets in $\mathcal{T}$ belong to $\mathcal{T}$.

Definition 10.2. Compactness
A topological space X is compact if every open cover of X has a finite subcover.

Definition 10.3. Connectedness
A topological space X is connected if it cannot be written as the union of two disjoint nonempty open sets. Equivalently, the only subsets of X that are both open and closed are $\varnothing$ and X.

Definition 10.4. Homeomorphism
A homeomorphism is a bijective continuous map whose inverse is also continuous. Two spaces are homeomorphic if they are the same from the viewpoint of topology.

Theorem 10.5. Brouwer Fixed-Point Theorem
Every continuous map from a closed ball in $\mathbb{R}^n$ to itself has at least one fixed point.

Theorem 10.6. Invariance of Domain
If $U \subseteq \mathbb{R}^n$ is open and $f : U \to \mathbb{R}^n$ is continuous and injective, then $f(U)$ is open and f is a homeomorphism from U onto $f(U)$.

Definition 10.7. Manifold
An n-dimensional topological manifold is a space that locally looks like $\mathbb{R}^n$. More precisely, each point has a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$.

Definition 10.8. Fundamental Group
The fundamental group $\pi_1(X, x_0)$ records loops in X based at x_0, up to continuous deformation. It is one of the basic algebraic invariants of a topological space.

Fact 10.9. Fundamental Groups of Familiar Spaces
The following basic examples are worth recognizing:

$$\pi_1(S^1) \cong \mathbb{Z}, \qquad \pi_1(S^n) \cong 0 \quad (n \geq 2), \qquad \pi_1(\mathbb{T}^2) \cong \mathbb{Z}^2.$$

Formula 10.10. Euler Characteristic of a Surface
For a triangulated closed surface, the Euler characteristic is

$$\chi = V - E + F.$$

For an orientable closed surface of genus g,

$$\chi = 2 - 2g.$$

Theorem 10.11. Classification of Closed Surfaces
Every connected closed surface is homeomorphic to either an orientable surface of genus $g \geq 0$ or a nonorientable surface obtained as a connected sum of $k \geq 1$ projective planes.

Theorem 10.12. Poincaré Conjecture
Every closed simply connected 3-manifold is homeomorphic to the 3-sphere S^3.

Remark. The Poincaré conjecture was proved by Grigori Perelman using Ricci flow. In a quiz setting, the key association is usually

$$\text{Poincaré conjecture} \longleftrightarrow \text{Perelman} \longleftrightarrow \text{Ricci flow.}$$

Theorem 10.13. Thurston Geometrization Theorem
Every closed orientable 3-manifold can be decomposed into pieces that admit one of the eight Thurston geometries.

Fact 10.14. The Eight Thurston Geometries
The eight model geometries are

$$S^3, \quad \mathbb{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}_2(\mathbb{R})}, \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Chapter 11

Miscellaneous Topics

Formula 11.1. Mercator Series for $\ln 2$
The alternating harmonic series satisfies

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2.$$

Remark. This follows from the Mercator series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

by substituting $x = 1$.

Formula 11.2. Basel Sum; Euler's Basel Formula
The Basel sum is

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}.$$

Equivalently,

$$\zeta(2) = \frac{\pi^2}{6}.$$

Remark. The problem of evaluating

$$\sum_{n=1}^{\infty}\frac{1}{n^2}$$

is called the Basel problem. Its famous solution was found by Euler.

Formula 11.3. Odd Part of the Basel Sum
The sum of reciprocal squares over odd positive integers is

$$\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}.$$

Idea. This is obtained by subtracting the even part from the Basel sum:

$$\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \sum_{n=1}^{\infty}\frac{1}{n^2} - \sum_{n=1}^{\infty}\frac{1}{(2n)^2} = \left(1 - \frac{1}{4}\right)\frac{\pi^2}{6} = \frac{\pi^2}{8}.$$

Formula 11.4. Alternating Basel Series; Dirichlet Eta Value at 2
The alternating Basel series satisfies

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}.$$

Equivalently,

$$\eta(2) = \frac{\pi^2}{12},$$

where $\eta(s)$ is the Dirichlet eta function.

Idea. The positive terms are the odd reciprocal squares and the negative terms are the even reciprocal squares. Hence

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^2} = \sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} - \sum_{n=1}^{\infty}\frac{1}{(2n)^2} = \frac{\pi^2}{8} - \frac{\pi^2}{24} = \frac{\pi^2}{12}.$$

Formula 11.5. Gregory–Leibniz Series; Leibniz Formula for π
The Gregory–Leibniz series is

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{2n-1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

Equivalently,

$$\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right).$$

Remark. This follows from the Taylor series

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$$

by substituting $x = 1$, since $\arctan 1 = \pi/4$.

Strategy. These values are useful because Science Quiz problems often hide them behind a simple transformation. For example,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right) = \frac{\ln 2}{2}.$$

Part II

Facts to Memorize

Chapter 12

Mathematical Prizes and Congresses

Science Quiz often asks for the short association between a prize, a congress, a laureate, and a mathematical achievement. This chapter records the facts that should be recognized quickly: not full biographies, but the names, years, keywords, and historical landmarks most likely to become quiz answers.

Fields Medal and ICM

The International Congress of Mathematicians, commonly abbreviated as ICM, is the most important international congress in mathematics. The Fields Medal is awarded by the International Mathematical Union at the ICM every four years. It is named after John Charles Fields. A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress.

Year	ICM location	Medalists	Quiz keywords
1936	Oslo, Norway	Lars Ahlfors; Jesse Douglas	Complex analysis, Riemann surfaces; minimal surfaces, Plateau problem.
1950	Cambridge, USA	Laurent Schwartz; Atle Selberg	Distribution theory; analytic number theory, sieve methods, zeta functions.
1954	Amsterdam, Netherlands	Kunihiko Kodaira; Jean-Pierre Serre	Complex geometry, algebraic geometry; algebraic topology, sheaf theory.
1958	Edinburgh, UK	Klaus Roth; René Thom	Diophantine approximation; cobordism, differential topology.
1962	Stockholm, Sweden	Lars Hörmander; John Milnor	Partial differential equations; differential topology, exotic spheres.
1966	Moscow, USSR	Michael Atiyah; Paul Cohen; Alexander Grothendieck; Stephen Smale	K-theory and index theorem; forcing and continuum hypothesis; modern algebraic geometry; generalized Poincaré conjecture.
1970	Nice, France	Alan Baker; Heisuke Hironaka; Serge Novikov; John G. Thompson	Transcendental number theory; resolution of singularities; topology and geometry; finite group theory.
1974	Vancouver, Canada	Enrico Bombieri; David Mumford	Number theory, complex analysis, minimal surfaces; moduli spaces and algebraic geometry.

Continued on the next page.

Year	ICM location	Medalists	Quiz keywords
1978	Helsinki, Finland	Pierre Deligne; Charles Fefferman; Grigory Margulis; Daniel Quillen	Weil conjectures; several complex variables; Lie groups and ergodic theory; algebraic K-theory.
1982	Warsaw, Poland	Alain Connes; William Thurston; Shing-Tung Yau	Noncommutative geometry; 3-manifolds and low-dimensional topology; Calabi conjecture and geometric analysis.
1986	Berkeley, USA	Simon Donaldson; Gerd Faltings; Michael Freedman	Gauge theory and 4-manifolds; Mordell conjecture and arithmetic geometry; topological 4-dimensional Poincaré conjecture.
1990	Kyoto, Japan	Vladimir Drinfeld; Vaughan Jones; Shigefumi Mori; Edward Witten	Quantum groups and Langlands program; Jones polynomial; minimal model program; mathematical physics and quantum field theory.
1994	Zurich, Switzerland	Jean Bourgain; Pierre-Louis Lions; Jean-Christophe Yoccoz; Efim Zelmanov	Analysis and Banach spaces; partial differential equations; dynamical systems; restricted Burnside problem.
1998	Berlin, Germany	Richard Borcherds; W. Timothy Gowers; Maxim Kontsevich; Curtis T. McMullen	Moonshine and automorphic forms; functional analysis and combinatorics; algebraic geometry and mathematical physics; complex dynamics.
2002	Beijing, China	Laurent Lafforgue; Vladimir Voevodsky	Langlands correspondence over function fields; motivic cohomology and $\mathbb{A}^1$-homotopy theory.
2006	Madrid, Spain	Andrei Okounkov; Grigori Perelman; Terence Tao; Wendelin Werner	Probability, representation theory, algebraic geometry; Ricci flow; PDE, harmonic analysis, additive combinatorics; SLE and Brownian motion.
2010	Hyderabad, India	Elon Lindenstrauss; Ngo Bao Chau; Stanislav Smirnov; Cédric Villani	Ergodic theory and number theory; Fundamental Lemma; conformal invariance; Boltzmann equation and optimal transport.
2014	Seoul, South Korea	Artur Avila; Manjul Bhargava; Martin Hairer; Maryam Mirzakhani	Dynamical systems; geometry of numbers; stochastic PDE; Riemann surfaces and moduli spaces.
2018	Rio de Janeiro, Brazil	Caucher Birkar; Alessio Figalli; Peter Scholze; Akshay Venkatesh	Fano varieties and minimal model program; optimal transport; p-adic geometry; number theory and homogeneous dynamics.
2022	Helsinki, Finland	Hugo Duminil-Copin; June Huh; James Maynard; Maryna Viazovska	Phase transitions in statistical physics; Hodge theory in combinatorics and matroids; prime numbers; sphere packing and the E_8 lattice.

Fact 12.1. Essential ICM and Fields Medal landmarks
The following associations are especially useful.

- ICM stands for International Congress of Mathematicians.
- IMU stands for International Mathematical Union.
- The Fields Medal was first awarded in 1936 at ICM Oslo.

- The first two Fields Medalists were Lars Ahlfors and Jesse Douglas.
- The first year with four Fields Medalists was 1966.
- Jean-Pierre Serre is the youngest Fields Medalist.
- Edward Witten was the first physicist to receive the Fields Medal.
- Andrew Wiles received a special IMU silver plaque in 1998 for his proof of Fermat's Last Theorem.
- Grigori Perelman declined the Fields Medal in 2006.
- Maryam Mirzakhani was the first woman Fields Medalist.
- Maryna Viazovska was the second woman Fields Medalist.
- ICM 2014 was held in Seoul, South Korea.
- ICM 2022 was held as a virtual congress; the prize ceremony was held in Helsinki, Finland.
- ICM 2026 is scheduled to be held in Philadelphia, USA.

Abel Prize

The Abel Prize is awarded by the Norwegian Academy of Science and Letters. It is named after Niels Henrik Abel and was first awarded in 2003. Unlike the Fields Medal, the Abel Prize has no age restriction. For Science Quiz, the Abel Prize is often asked through the pattern

$$\text{year} \quad \longleftrightarrow \quad \text{laureate} \quad \longleftrightarrow \quad \text{famous theorem or area.}$$

Year	Laureate(s)	Quiz keywords
2003	Jean-Pierre Serre	Algebraic topology, algebraic geometry, number theory; first Abel Prize laureate.
2004	Michael Atiyah; Isadore Singer	Atiyah–Singer index theorem.
2005	Peter D. Lax	Partial differential equations and applied mathematics.
2006	Lennart Carleson	Harmonic analysis; convergence of Fourier series.
2007	Srinivasa S. R. Varadhan	Probability theory; large deviations.
2008	John G. Thompson; Jacques Tits	Finite groups, classification of finite simple groups; buildings and group theory.
2009	Mikhail Gromov	Geometry, geometric group theory, symplectic geometry.
2010	John Tate	Number theory and arithmetic geometry.
2011	John Milnor	Topology, geometry, dynamics; exotic spheres.
2012	Endre Szemerédi	Discrete mathematics and theoretical computer science; Szemerédi's theorem.
2013	Pierre Deligne	Algebraic geometry; Weil conjectures.
2014	Yakov Sinai	Dynamical systems, ergodic theory, mathematical physics.
2015	John F. Nash Jr.; Louis Nirenberg	Nonlinear PDE, geometry, Nash embedding theorem.

Continued on the next page.

Year	Laureate(s)	Quiz keywords
2016	Andrew Wiles	Fermat's Last Theorem; modularity of elliptic curves.
2017	Yves Meyer	Wavelet theory; applications from signal compression to gravitational-wave analysis.
2018	Robert P. Langlands	Langlands program; representation theory and number theory.
2019	Karen Uhlenbeck	Geometric analysis and gauge theory; first woman Abel Prize laureate.
2020	Hillel Furstenberg; Gregory Margulis	Probability, dynamics, ergodic theory, Lie groups.
2021	László Lovász; Avi Wigderson	Discrete mathematics, theoretical computer science, complexity theory.
2022	Dennis P. Sullivan	Topology, geometric topology, dynamical systems.
2023	Luis A. Caffarelli	Nonlinear partial differential equations, free-boundary problems, Monge–Ampère equation.
2024	Michel Talagrand	Probability theory, functional analysis, spin glasses, concentration inequalities.
2025	Masaki Kashiwara	Algebraic analysis, D-modules, representation theory, crystal bases.
2026	Gerd Faltings	Arithmetic geometry; Mordell conjecture, Lang's conjectures, Diophantine geometry.

Fact 12.2. Essential Abel Prize landmarks
The following associations are especially likely to be useful.

- Abel Prize ↔ Norwegian Academy of Science and Letters.
- Abel Prize ↔ Niels Henrik Abel.
- First Abel Prize laureate ↔ Jean-Pierre Serre.
- First woman Abel Prize laureate ↔ Karen Uhlenbeck.
- Andrew Wiles ↔ Abel Prize 2016 ↔ Fermat's Last Theorem.
- Yves Meyer ↔ Abel Prize 2017 ↔ wavelets.
- Robert Langlands ↔ Abel Prize 2018 ↔ Langlands program.
- Gerd Faltings ↔ Fields Medal 1986 and Abel Prize 2026.

Wolf Prize in Mathematics

The Wolf Prize is awarded by the Wolf Foundation. In the sciences, the Wolf Prize is awarded in Medicine, Agriculture, Mathematics, Chemistry, and Physics. It is especially important historically because many Wolf Prize in Mathematics laureates later received, or had already received, other major mathematical prizes. Unlike the Fields Medal, it is not restricted by age.

Year	Laureate(s)	Quiz keywords
1978	Israel Gelfand; Carl L. Siegel	Representation theory, functional analysis; number theory, several complex variables.

Continued on the next page.

Year	Laureate(s)	Quiz keywords
1979	Jean Leray; André Weil	PDE and algebraic topology; algebraic geometry and number theory.
1980	Henri Cartan; Andrey Kolmogorov	Several complex variables and algebraic topology; probability theory and dynamical systems.
1981	Lars Ahlfors; Oscar Zariski	Complex analysis and Riemann surfaces; algebraic geometry.
1982	Mark Krein; Hassler Whitney	Functional analysis; differential topology and Whitney embedding.
1983	Shiing-Shen Chern	Global differential geometry, Chern classes.
1984	Paul Erdős	Combinatorics, number theory, probabilistic method.
1985	Kunihiko Kodaira; Hans Lewy	Complex geometry; partial differential equations.
1986	Samuel Eilenberg; Atle Selberg	Algebraic topology and category theory; analytic number theory and Selberg trace formula.
1987	Kiyoshi Itô; Peter D. Lax	Stochastic calculus; partial differential equations and applied mathematics.
1988	Lars Hörmander; Friedrich Hirzebruch	Partial differential equations; topology, characteristic classes, Hirzebruch–Riemann–Roch.
1989	Alberto Calderón; John Milnor	Harmonic analysis and singular integrals; topology and exotic spheres.
1990	Ennio De Giorgi; Ilya Piatetski-Shapiro	Minimal surfaces and regularity theory; automorphic forms.
1992	Lennart Carleson; John G. Thompson	Harmonic analysis; finite group theory.
1993	Mikhail Gromov; Jacques Tits	Geometry and geometric group theory; buildings and group theory.
1995/96	Robert Langlands; Andrew Wiles	Langlands program; Fermat's Last Theorem.
1996/97	Joseph Keller; Yakov Sinai	Applied mathematics and waves; dynamical systems and mathematical physics.
1999	László Lovász; Elias M. Stein	Discrete mathematics and algorithms; harmonic analysis.
2000	Raoul Bott; Jean-Pierre Serre	Topology and geometry; algebraic topology, algebraic geometry, number theory.
2001	Vladimir Arnold; Saharon Shelah	Dynamical systems and singularity theory; mathematical logic and set theory.
2002/03	Mikio Sato; John Tate	Algebraic analysis and hyperfunctions; arithmetic geometry.
2005	Gregory Margulis; Serge Novikov	Lie groups, ergodic theory, discrete subgroups; topology and geometry.
2006/07	Stephen Smale; Hillel Furstenberg	Differential topology and dynamical systems; ergodic theory and probability.
2008/09	Pierre Deligne; Phillip Griffiths; David Mumford	Algebraic geometry, Hodge theory, moduli spaces.
2010	Shing-Tung Yau; Dennis Sullivan	Geometric analysis; topology and dynamics.
2012	Michael Aschbacher; Luis Caffarelli	Finite groups; nonlinear PDE and free-boundary problems.
2013	George Mostow; Michael Artin	Rigidity, geometry, Lie groups; algebraic geometry.

Continued on the next page.

Year	Laureate(s)	Quiz keywords
2014	Peter Sarnak	Number theory, analysis, geometry, automorphic forms.
2015	James Arthur	Trace formula, automorphic representations.
2017	Richard Schoen; Charles Fefferman	Geometric analysis; analysis and partial differential equations.
2018	Alexander Beilinson; Vladimir Drinfeld	Geometric representation theory, mathematical physics, Langlands program.
2019	Jean-François Le Gall; Gregory Lawler	Probability theory, random walks, Brownian motion, SLE.
2020	Simon Donaldson; Yakov Eliashberg	Differential geometry, topology, gauge theory; symplectic and contact topology.
2022	George Lusztig	Representation theory and related areas.
2023	Ingrid Daubechies	Wavelet theory, time-frequency analysis, signal processing.
2024	Adi Shamir; Noga Alon	Mathematical cryptography; combinatorics, graph theory, theoretical computer science.

Fact 12.3. Essential Wolf Prize landmarks
The following associations are the most useful ones to memorize first.

- Wolf Prize ↔ Wolf Foundation.
- Wolf Prize in Mathematics ↔ no Fields-style age restriction.
- Paul Erdős ↔ Wolf Prize 1984 ↔ combinatorics and probabilistic method.
- Andrew Wiles and Robert Langlands shared the 1995/96 Wolf Prize in Mathematics.
- Ingrid Daubechies ↔ Wolf Prize 2023 ↔ wavelets.
- Adi Shamir and Noga Alon ↔ Wolf Prize 2024 ↔ cryptography and combinatorics.

Chapter 13

Famous Problems and Conjectures

Famous problems are usually not tested by asking for complete proofs. They are tested through the following information: the standard statement, the person or group attached to the problem, whether it is solved, and the main idea or keyword attached to the solution. The goal of this chapter is to make those associations immediate.

Millennium Prize Problems

The Clay Mathematics Institute announced the seven Millennium Prize Problems in 2000. Each problem carries a prize of one million US dollars. Among the seven, only the Poincaré conjecture has been solved.

Problem	Standard mathematical form	Status	Quiz anchors
Poincaré Conjecture	Every closed simply connected 3-manifold is homeomorphic to S^3.	Solved.	Poincaré; Perelman; Ricci flow.
Riemann Hypothesis	Every nontrivial zero of $\zeta(s)$ has real part 1/2.	Open.	Riemann; zeta function; prime numbers.
P versus NP	Is $P = NP$?	Open.	Cook, Karp; computational complexity.
Birch and Swinnerton-Dyer Conjecture	For an elliptic curve E, the rank of $E(\mathbb{Q})$ equals the order of vanishing of $L(E, s)$ at $s = 1$.	Open.	Elliptic curves; L-functions.
Hodge Conjecture	Rational Hodge classes on smooth projective complex varieties should come from algebraic cycles.	Open.	Hodge theory; algebraic geometry.
Navier–Stokes Existence and Smoothness	Smooth initial data for the 3-dimensional incompressible Navier–Stokes equations should have global smooth solutions, or one must find a breakdown.	Open.	Fluid mechanics; PDE; turbulence.
Yang–Mills Existence and Mass Gap	Construct quantum Yang–Mills theory on $\mathbb{R}^4$ for compact simple gauge groups and prove a positive mass gap.	Open.	Gauge theory; quantum field theory; mass gap.

Theorem 13.1. Poincaré Conjecture
Every closed simply connected 3*-manifold is homeomorphic to the* 3*-sphere* S^3*.*

Remark. Henri Poincaré posed the problem in the early twentieth century. Grigori Perelman proved it using Ricci flow and ideas from Hamilton's program. The key Science Quiz association is

$$\text{Poincaré conjecture} \quad \longleftrightarrow \quad \text{Perelman} \quad \longleftrightarrow \quad \text{Ricci flow.}$$

Hypothesis 13.2. Riemann Hypothesis
Let

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

be the Riemann zeta function, initially defined for $\mathrm{Re}(s) > 1$ *and continued meromorphically. The Riemann Hypothesis states that every nontrivial zero of* $\zeta(s)$ *has real part* $1/2$*.*

Idea. The zeta function is connected to prime numbers by Euler's product

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Thus RH is a statement about the hidden regularity of the distribution of prime numbers.

Open Problem 13.3. P versus NP
Determine whether

$$P = NP.$$

Here P is the class of decision problems solvable in polynomial time, and NP is the class of decision problems whose proposed solutions can be verified in polynomial time.

Remark. The usual slogan is: can every solution that can be checked quickly also be found quickly? The keywords are Cook, Karp, NP-completeness, and theoretical computer science.

Conjecture 13.4. Birch and Swinnerton-Dyer Conjecture
For an elliptic curve E *over* $\mathbb{Q}$*, the rank of the abelian group* $E(\mathbb{Q})$ *equals the order of vanishing of its* L*-function* $L(E, s)$ *at* $s = 1$*.*

Remark. The problem connects rational points on elliptic curves with analytic behavior of L-functions. The phrase "rank equals order of vanishing" is the main quiz memory.

Conjecture 13.5. Hodge Conjecture
On a smooth projective complex variety, every rational cohomology class of type (p, p) *should be a rational linear combination of classes of algebraic cycles.*

Remark. This is one of the deepest bridges between topology and algebraic geometry. The short memory is

$$\text{Hodge classes} \quad \longleftrightarrow \quad \text{algebraic cycles.}$$

Open Problem 13.6. Navier–Stokes Existence and Smoothness
For the 3-dimensional incompressible Navier–Stokes equations, determine whether smooth initial data always lead to global smooth solutions, or whether finite-time singularities can occur.

Remark. This is a PDE problem motivated by fluid mechanics. The quiz keywords are incompressible fluid, global regularity, singularity, and turbulence.

Open Problem 13.7. Yang–Mills Existence and Mass Gap
Construct a mathematically rigorous quantum Yang–Mills theory on $\mathbb{R}^4$ for every compact simple gauge group and prove that it has a positive mass gap.

Remark. The problem comes from quantum field theory. In quiz form, the key words are gauge theory, quantum field theory, and mass gap.

Classical Problems and Conjectures

The following problems are not all open problems. Some are solved theorems, but they remain famous because of their history, statements, or solution methods.

Problem	Standard form	People and status	Quiz anchor
Fermat's Last Theorem	$x^n + y^n = z^n$ has no positive integer solutions for $n \geq 3$.	Fermat; proved by Wiles with Taylor–Wiles.	Elliptic curves and modular forms.
Four Color Theorem	Every planar map can be colored with at most four colors.	Appel–Haken; later Robertson–Sanders–Seymour–Thomas.	First famous computer-assisted proof.
Kepler Conjecture	The densest packing of congruent spheres in $\mathbb{R}^3$ is the face-centered cubic packing.	Kepler; proved by Hales.	Sphere packing; Flyspeck project.
Thurston Geometrization Conjecture	Closed orientable 3-manifolds decompose into pieces modeled on eight geometries.	Thurston; completed through Perelman's work.	Eight Thurston geometries.
Continuum Hypothesis	There is no cardinality strictly between $\lvert\mathbb{N}\rvert$ and $\lvert\mathbb{R}\rvert$.	Cantor; independent of ZFC by Gödel and Cohen.	Forcing; independence.
Hilbert's Problems	A list of 23 problems proposed in 1900.	Hilbert; many solved, some open or independent.	Roadmap for twentieth-century mathematics.
Goldbach Conjecture	Every even integer greater than 2 is a sum of two primes.	Goldbach; open.	Additive number theory.
Twin Prime Conjecture	There are infinitely many primes p such that $p + 2$ is also prime.	Open; bounded gaps by Zhang, Maynard, Tao, Polymath.	Prime gaps.
Collatz Conjecture	Iterating $n \mapsto n/2$ for even n and $n \mapsto 3n + 1$ for odd n eventually reaches 1.	Collatz; open.	Simple statement, hard dynamics.
abc Conjecture	If $a + b = c$ and a, b, c are coprime, then c is controlled by the radical of abc, up to ε.	Masser–Oesterlé; open in the standard sense.	Radical, Diophantine equations.

Theorem 13.8. Fermat's Last Theorem
For every integer $n \geq 3$, there do not exist positive integers x, y, z such that

$$x^n + y^n = z^n.$$

Remark. Fermat wrote that he had a marvelous proof, but no proof appeared in his margin. Andrew Wiles proved the theorem through the modularity of semistable elliptic curves. The fast association is

$$\text{Fermat's Last Theorem} \quad \longleftrightarrow \quad \text{Wiles} \quad \longleftrightarrow \quad \text{elliptic curves and modular forms.}$$

Theorem 13.9. Four Color Theorem
Every planar map can be colored with at most four colors so that any two adjacent regions have different colors.

Remark. The theorem was proved by Kenneth Appel and Wolfgang Haken in 1976 with extensive computer assistance. This made it one of the most famous early examples of a computer-assisted proof.

Theorem 13.10. Kepler Conjecture
The maximum density of a packing of congruent spheres in $\mathbb{R}^3$ *is attained by the face-centered cubic packing and equals*

$$\frac{\pi}{3\sqrt{2}}.$$

Remark. Thomas Hales proved the Kepler conjecture using a large computer-assisted argument. The Flyspeck project later produced a formal verification of the proof. In higher-dimensional sphere packing, Maryna Viazovska proved the optimality of the E_8 lattice in dimension 8.

Theorem 13.11. Thurston Geometrization Theorem
Every closed orientable 3*-manifold can be decomposed into pieces that admit one of the eight Thurston geometries:*

$$S^3,\ E^3,\ H^3,\ S^2 \times \mathbb{R},\ H^2 \times \mathbb{R},\ \widetilde{\mathrm{SL}_2(\mathbb{R})},\ \mathrm{Nil},\ \mathrm{Sol}.$$

Remark. The Poincaré conjecture is a special case of the geometrization program. The 2018 Science Quiz asked for the number of Thurston geometries; the answer is 8.

Hypothesis 13.12. Continuum Hypothesis
There is no set whose cardinality is strictly between the cardinality of the natural numbers and the cardinality of the real numbers. Equivalently,

$$2^{\aleph_0} = \aleph_1.$$

Remark. Kurt Gödel proved that CH cannot be disproved from ZFC if ZFC is consistent. Paul Cohen proved that CH cannot be proved from ZFC if ZFC is consistent. Cohen introduced forcing and received the Fields Medal in 1966.

Fact 13.13. Hilbert's Problems
David Hilbert presented 23 problems at the 1900 International Congress of Mathematicians in Paris. They shaped much of twentieth-century mathematics. Useful anchors include:

- Hilbert's first problem $\leftrightarrow$ continuum hypothesis.
- Hilbert's second problem $\leftrightarrow$ consistency of arithmetic.
- Hilbert's tenth problem $\leftrightarrow$ Diophantine equations; negative solution by Matiyasevich, building on Davis–Putnam–Robinson.

Open Problem 13.14. Goldbach Conjecture
Every even integer greater than 2 is a sum of two prime numbers.

Remark. Goldbach is one of the easiest famous open problems to state. It belongs to additive number theory. The weak Goldbach conjecture, concerning sums of three odd primes, was proved by Harald Helfgott.

Open Problem 13.15. Twin Prime Conjecture
There are infinitely many primes p such that

$$p + 2$$

is also prime.

Remark. The modern quiz keywords are bounded gaps between primes, Yitang Zhang, James Maynard, Terence Tao, and the Polymath project. These results do not prove the twin prime conjecture, but they prove that infinitely many prime pairs occur within some fixed bounded distance.

Open Problem 13.16. Collatz Conjecture
Start with a positive integer n. If n is even, replace it by $n/2$. If n is odd, replace it by $3n + 1$. The Collatz conjecture states that repeated iteration always eventually reaches 1.

Remark. This problem is famous because the rule is elementary but the global behavior is still unknown. It is also called the $3n + 1$ problem.

Conjecture 13.17. *abc* **Conjecture**
For every $\varepsilon > 0$*, there exists a constant* K_ε *such that, whenever* a, b, c *are pairwise coprime positive integers satisfying* $a + b = c$*, we have*

$$c < K_\varepsilon \operatorname{rad}(abc)^{1+\varepsilon},$$

where $\operatorname{rad}(n)$ *is the product of the distinct prime divisors of* n*.*

Remark. The conjecture is powerful because it predicts that the additive relation $a + b = c$ strongly restricts the repeated prime factors of *abc*. The keyword is "radical."

Modern Quiz-Relevant Conjecture Clusters

Some recent Science Quiz-style questions are built around a mathematician and the conjectures attached to that person's work. The most important example for current preparation is June Huh.

Fact 13.18. June Huh and combinatorial log-concavity
June Huh's work connects combinatorics with algebraic geometry and Hodge theory. The most useful associations are:

- Read's conjecture $\leftrightarrow$ chromatic polynomials.
- Rota's conjecture $\leftrightarrow$ log-concavity for matroids.
- Mason's conjecture $\leftrightarrow$ independent sets of matroids.
- Hodge theory $\leftrightarrow$ the geometric method behind these combinatorial log-concavity results.

In a quiz, if June Huh appears with a list of conjectures, the odd one out is often a famous conjecture from a different field, such as the Poincaré conjecture.

Part III

Mathematics in 2026

Chapter 14

Mathematics in 2026

This part is reserved for mathematical events, awards, anniversaries, and recent developments that are especially relevant to the 2026 Science Quiz season. Unlike the other parts of this book, this part is meant to be replaced every year.

Update Checklist

- Major international mathematical events held in 2026.
- Prize winners and award announcements from 2026.
- Important anniversaries of mathematicians, theorems, or famous mathematical events.
- Recent breakthroughs or widely discussed results that may appear in quiz-style questions.
- POSTECH–KAIST Science War specific trends from the current season.

Working Notes

Add the 2026-specific memorization items here. When preparing the next edition, replace this file by `Mathematicsin2027.tex` and update the corresponding input line in the main file.

Part IV

Past Problems

Problem 2011.1. What is the name of the international mathematical congress, held in South Korea in 2014, at which the Fields Medal is awarded?

Solution. The Fields Medal is awarded at the International Congress of Mathematicians, commonly abbreviated as ICM. The 2014 ICM was held in Seoul, South Korea.

Therefore, the answer is $\boxed{\text{International Congress of Mathematicians, or ICM}}$.

$\square$

Problem 2011.2. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. If

$$\mathbf{AB} - \mathbf{I} = \mathbf{0},$$

where $\mathbf{I}$ is the identity matrix, what is the value of

$$\mathbf{BA} - \mathbf{I}?$$

Solution. The condition

$$\mathbf{AB} - \mathbf{I} = \mathbf{0}$$

means that

$$\mathbf{AB} = \mathbf{I}.$$

Since $\mathbf{A}$ and $\mathbf{B}$ are square matrices, this implies that $\mathbf{B}$ is the inverse of $\mathbf{A}$. Hence

$$\mathbf{BA} = \mathbf{I}.$$

Therefore,

$$\mathbf{BA} - \mathbf{I} = \mathbf{I} - \mathbf{I} = \mathbf{0}.$$

Thus the answer is $\boxed{\mathbf{0}}$.

$\square$

Problem 2011.3. What is the value obtained by differentiating $(1 + x)^{10}$ three times and then substituting $x = 0$?

Solution. We compute the third derivative:

$$\frac{d^3}{dx^3}(1 + x)^{10} = 10 \cdot 9 \cdot 8(1 + x)^7.$$

Substituting $x = 0$, we get

$$10 \cdot 9 \cdot 8(1 + 0)^7 = 720.$$

Therefore, the answer is $\boxed{720}$.

$\square$

Problem 2011.4. Give at least two real-valued functions defined on $\mathbb{R}$ that remain unchanged after being differentiated twice. In other words, give at least two functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f'' = f.$$

Solution. Two simple examples are

$$f(x) = e^x \qquad \text{and} \qquad g(x) = e^{-x}.$$

Indeed,

$$\frac{d^2}{dx^2}e^x = e^x$$

and

$$\frac{d^2}{dx^2}e^{-x} = e^{-x}.$$

Therefore, both functions remain unchanged after being differentiated twice.

Thus two possible answers are $\boxed{e^x \text{ and } e^{-x}}$.

□

Problem 2011.5. Let $\mathbf{A}$ be a 5×5 matrix such that every entry in the i-th row is equal to i. Find the sum of the five eigenvalues of $\mathbf{A}$, counted with multiplicity.

Solution. The sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to its trace.

In this matrix, the diagonal entries are

$$1, \quad 2, \quad 3, \quad 4, \quad 5.$$

Therefore,

$$\mathrm{tr}(\mathbf{A}) = 1 + 2 + 3 + 4 + 5 = 15.$$

Hence the sum of the five eigenvalues of $\mathbf{A}$, counted with multiplicity, is $\boxed{15}$.

□

Problem 2011.6. A train takes 22 seconds to completely cross an 82m bridge. When its speed is doubled, it takes 50 seconds to completely cross a 706m bridge. What is the length of the train?

Solution. Let L be the length of the train, and let v be its original speed in meters per second.

To completely cross a bridge, the train must travel the length of the bridge plus its own length. Hence

$$\frac{L + 82}{v} = 22.$$

When the speed is doubled, the train takes 50 seconds to completely cross a 706m bridge, so

$$\frac{L + 706}{2v} = 50.$$

Equivalently,

$$L + 82 = 22v, \qquad L + 706 = 100v.$$

Subtracting the first equation from the second gives

$$624 = 78v,$$

so

$$v = 8.$$

Therefore,

$$L + 82 = 22 \cdot 8 = 176,$$

and hence

$$L = 94.$$

Thus the length of the train is $\boxed{94\text{m}}$.

□

Problem 2011.7. Form an integer using the digits

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Each of these digits must be used at least once, and digits may be repeated. What is the smallest such integer that is divisible by all of

$$1, 2, 3, 4, 6, 7, 8, 9?$$

Solution. The number must be divisible by

$$\mathrm{lcm}(1, 2, 3, 4, 6, 7, 8, 9) = 504.$$

Thus it must be divisible by 7, 8, and 9.

If each digit is used exactly once, then the sum of the digits is

$$1 + 2 + 3 + 4 + 6 + 7 + 8 + 9 = 40,$$

which is not divisible by 9. Hence no 8-digit number works.

If we use exactly one extra digit d, then the digit sum becomes $40 + d$. For divisibility by 9, we need

$$40 + d \equiv 0 \pmod 9,$$

so

$$d \equiv 5 \pmod 9.$$

But the digit 5 is not allowed. Hence no 9-digit number works.

Therefore, the answer must have at least 10 digits. For a 10-digit number, we need to add two extra digits. Their sum must be congruent to 5 modulo 9. Among the allowed digits, the possible extra pairs are

$$(1, 4), \qquad (2, 3), \qquad (6, 8), \qquad (7, 7).$$

To make the number as small as possible, we should try the lexicographically smallest multiset first, namely

$$1, 1, 2, 3, 4, 4, 6, 7, 8, 9.$$

With these digits, the smallest possible beginning is

$$112344.$$

It remains to arrange the last four digits $6, 7, 8, 9$ so that the whole number is divisible by 504.

Since the digit sum is already divisible by 9, it remains to check divisibility by 7 and 8. The last three digits must be divisible by 8. Among the permutations of three of the digits $6, 7, 8, 9$, the possible last three digits divisible by 8 are

$$768, \qquad 896, \qquad 968, \qquad 976.$$

Hence the only candidates with prefix 112344 are

$$1123449768, \qquad 1123447896, \qquad 1123447968, \qquad 1123448976.$$

Checking divisibility by 7, only

$$1123449768$$

is divisible by 7. Indeed,

$$1123449768 = 504 \cdot 2229067.$$

Therefore it is divisible by all of

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Thus the smallest possible integer is $\boxed{1123449768}$.

$\square$

Problem 2011.8. What is the greatest integer less than or equal to

$$-\frac{1}{\ln\left(\frac{365}{366}\right)}?$$

Hint. Use the Taylor approximation of $\ln(1-x)$.

Solution. We write

$$\frac{365}{366} = 1 - \frac{1}{366}.$$

Using

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots,$$

with $x = \dfrac{1}{366}$, we get

$$-\ln\left(\frac{365}{366}\right) = \frac{1}{366} + \frac{1}{2 \cdot 366^2} + \frac{1}{3 \cdot 366^3} + \cdots.$$

Hence

$$\frac{1}{366} < -\ln\left(\frac{365}{366}\right) < \frac{1}{365}.$$

Taking reciprocals gives

$$365 < -\frac{1}{\ln\left(\frac{365}{366}\right)} < 366.$$

Therefore, the greatest integer less than or equal to the given number is $\boxed{365}$.

$\square$

Problem 2011.9. Evaluate the integral

$$\int_0^{\pi/2} \frac{1}{1 + \tan^{\sqrt{2}} t}\, dt.$$

Hint. Use the substitution $x = \dfrac{\pi}{2} - t$.

Solution. Let

$$I = \int_0^{\pi/2} \frac{1}{1 + \tan^{\sqrt{2}} t}\, dt.$$

Use the substitution

$$x = \frac{\pi}{2} - t.$$

Then

$$\tan t = \tan\left(\frac{\pi}{2} - x\right) = \cot x = \frac{1}{\tan x}.$$

Hence

$$I = \int_0^{\pi/2} \frac{1}{1 + \cot^{\sqrt{2}} x}\, dx = \int_0^{\pi/2} \frac{\tan^{\sqrt{2}} x}{1 + \tan^{\sqrt{2}} x}\, dx.$$

Adding this expression to the original expression for I, we obtain

$$2I = \int_0^{\pi/2} \left(\frac{1}{1+\tan^{\sqrt{2}} x} + \frac{\tan^{\sqrt{2}} x}{1+\tan^{\sqrt{2}} x} \right) dx = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2}.$$

Therefore,

$$I = \frac{\pi}{4}.$$

Thus the answer is $\boxed{\frac{\pi}{4}}$.

□

Problem 2011.10. This French mathematician was born on December 8, 1865, and died on October 17, 1963, living to the age of 97. He is well known for his proof of the prime number theorem, which states that the number of primes less than or equal to x is asymptotically approximated by $\frac{x}{\ln x}$ for large x. He was also personally connected to the Dreyfus affair and became a strong supporter of Zionism.

Who is this mathematician?

Hint. The following clues are useful:

(a) The Cauchy–__________ theorem gives the radius of convergence of a power series.

(b) A __________ matrix is a square matrix whose entries are all $+1$ or -1, and such matrices are also used in error-correcting codes.

Solution. The mathematician described in the problem is Jacques Hadamard.

The first hint refers to the Cauchy–Hadamard theorem, which gives the radius of convergence of a power series. The second hint refers to a Hadamard matrix, a square matrix whose entries are all $+1$ or -1 and whose rows are mutually orthogonal.

Hadamard is also famous for proving the prime number theorem in 1896, independently of Charles Jean de la Vallée Poussin. Therefore, the answer is $\boxed{\text{Jacques Hadamard}}$.

□

Problem 2012.1. This British mathematical genius died of cyanide poisoning shortly before his 42nd birthday. A famous account says that he died after eating an apple laced with cyanide. Who was this mathematician?

Solution. The mathematician described in the problem is Alan Turing.

Turing was a British mathematician, logician, and computer scientist. He made fundamental contributions to computability theory through the concept of the Turing machine, and he also played a crucial role in codebreaking during World War II.

Turing died in 1954 from cyanide poisoning, shortly before his 42nd birthday. The story involving an apple laced with cyanide is widely known, although some details of the event remain historically uncertain.

Therefore, the answer is $\boxed{\text{Alan Turing}}$.

□

Problem 2012.2. Evaluate the infinite series

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots .$$

Solution. The given series is

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

This is the famous Basel problem. Euler proved that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore, the value of the series is $\boxed{\frac{\pi^2}{6}}$.

□

Problem 2012.3. Who was the mathematician who first proved the law of quadratic reciprocity and discovered more than seven proofs of it?

Solution. The law of quadratic reciprocity is one of the central theorems in elementary number theory. It describes a deep symmetry between the solvability of

$$x^2 \equiv p \pmod{q} \qquad \text{and} \qquad x^2 \equiv q \pmod{p}$$

for odd primes p and q.

Carl Friedrich Gauss gave the first complete proof of the law of quadratic reciprocity. He regarded it as one of the jewels of number theory and found several different proofs of the theorem.

Therefore, the answer is $\boxed{\text{Carl Friedrich Gauss}}$.

□

Problem 2012.4. Let $\mathbf{A}$ be a 10×10 matrix such that every entry in the k-th row is equal to k. Find the sum of the eigenvalues of $\mathbf{A}$, counted with multiplicity.

Solution. The sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to its trace.

In this matrix, the diagonal entries are

$$1, \quad 2, \quad 3, \quad \ldots, \quad 10.$$

Therefore,

$$\operatorname{tr}(\mathbf{A}) = 1 + 2 + 3 + \cdots + 10 = \frac{10 \cdot 11}{2} = 55.$$

Hence the sum of the eigenvalues of $\mathbf{A}$, counted with multiplicity, is $\boxed{55}$.

□

Problem 2012.5. This philosopher was born as the youngest son of a wealthy Austrian steel magnate who supported many artists. He showed talent in several fields, especially mathematics and philosophy. His work had a major influence on logical positivism and on the philosophy of language, and it also influenced the humanities, social sciences, and the arts. His later philosophy is especially known for emphasizing ordinary language use.

Who is this person?

Solution. The person described in the problem is Ludwig Wittgenstein.

Wittgenstein was born into a wealthy Austrian family and studied logic, mathematics, and philosophy. His early work, especially the *Tractatus Logico-Philosophicus*, strongly influenced logical positivism and the Vienna Circle.

Later, in works such as *Philosophical Investigations*, Wittgenstein emphasized the importance of ordinary language and argued that the meaning of a word is closely connected to its use in language.

Therefore, the answer is $\boxed{\text{Ludwig Wittgenstein}}$.

□

Problem 2012.6. Re

Solution. Re

□

Problem 2014.1. This twentieth-century German mathematician is well known for a theorem stating that symmetries of Hamiltonian dynamical systems correspond to conservation laws. What is the name of this mathematician?

Solution. The theorem described in the problem is Noether's theorem. It states, roughly speaking, that continuous symmetries of a physical system correspond to conserved quantities. For example, time-translation symmetry corresponds to conservation of energy, and spatial-translation symmetry corresponds to conservation of momentum.

The mathematician associated with this theorem is Emmy Noether, one of the most important algebraists of the twentieth century.

Therefore, the answer is $\boxed{\text{Emmy Noether}}$.

□

Problem 2014.2. In the famous Chinese mathematical classic *The Nine Chapters on the Mathematical Art*, often compared with Euclid's *Elements* as a foundational mathematical classic, the value of π appears as 3. Later, Liu Hui, who wrote a commentary on *The Nine Chapters*, obtained the approximation 3.14 by refining the method of inscribed polygons. Which ancient Chinese mathematician further refined this method and obtained the approximations

$$\frac{355}{113} \quad \text{and} \quad \frac{22}{7}$$

for π?

Solution. The mathematician described in the problem is Zu Chongzhi.

Liu Hui improved the approximation of π by using polygons with many sides. Zu Chongzhi later obtained the very accurate bounds

$$3.1415926 < \pi < 3.1415927,$$

and gave two famous rational approximations:

$$\frac{22}{7} \quad \text{and} \quad \frac{355}{113}.$$

The approximation $\frac{355}{113}$ is especially accurate and is traditionally known as the Milü, or "accurate ratio."

Therefore, the answer is $\boxed{\text{Zu Chongzhi}}$.

□

Problem 2014.3. Consider the 2014×2014 matrix whose diagonal entries are all 0 and whose off-diagonal entries are all 1. Find all eigenvalues of this matrix.

Solution. Let $\mathbf{A}$ be the given matrix, and let $\mathbf{J}$ be the 2014×2014 matrix whose entries are all 1. Then

$$\mathbf{A} = \mathbf{J} - \mathbf{I}.$$

The matrix $\mathbf{J}$ has eigenvalue 2014 for the vector

$$\mathbf{1} = (1, 1, \ldots, 1)^T,$$

since

$$\mathbf{J1} = 2014\mathbf{1}.$$

Therefore,

$$\mathbf{A1} = (\mathbf{J} - \mathbf{I})\mathbf{1} = 2014\mathbf{1} - \mathbf{1} = 2013\mathbf{1}.$$

Hence 2013 is an eigenvalue of $\mathbf{A}$.

Now take any vector $\mathbf{v} = (v_1, \ldots, v_{2014})^T$ such that

$$v_1 + \cdots + v_{2014} = 0.$$

Then $\mathbf{Jv} = \mathbf{0}$, so

$$\mathbf{Av} = (\mathbf{J} - \mathbf{I})\mathbf{v} = -\mathbf{v}.$$

Thus -1 is an eigenvalue. The subspace of vectors whose coordinates sum to 0 has dimension 2013, so the multiplicity of -1 is 2013.

Therefore, the eigenvalues are

$$\boxed{2013 \text{ with multiplicity } 1, \qquad -1 \text{ with multiplicity } 2013}.$$

$\square$

Problem 2014.4. Let $m \geq 3$ and $n \geq 3$. Suppose that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq \mathbb{R}^m, \qquad \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} \subseteq \mathbb{R}^n.$$

Assume that

$$V = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$$

is a 2-dimensional subspace of $\mathbb{R}^m$, and that

$$W = \operatorname{span}\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$$

is a 3-dimensional subspace of $\mathbb{R}^n$. What is the rank of the $m \times n$ matrix

$$\mathbf{A} = \mathbf{v}_1\mathbf{w}_1^T + \mathbf{v}_2\mathbf{w}_2^T + \mathbf{v}_3\mathbf{w}_3^T?$$

Solution. Let

$$\mathbf{P} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{pmatrix}, \qquad \mathbf{Q} = \begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 \end{pmatrix}.$$

Then

$$\mathbf{A} = \mathbf{P}\mathbf{Q}^{\mathsf{T}}.$$

Since $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ span a 2-dimensional subspace, we have

$$\operatorname{rk}(\mathbf{P}) = 2.$$

Since $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ span a 3-dimensional subspace, they are linearly independent, and hence

$$\operatorname{rk}(\mathbf{Q}) = 3.$$

Therefore, the linear map $\mathbf{Q}^{\mathsf{T}} : \mathbb{R}^n \to \mathbb{R}^3$ is surjective.

It follows that the image of $\mathbf{A} = \mathbf{P}\mathbf{Q}^{\mathsf{T}}$ is

$$\operatorname{im}(\mathbf{A}) = \mathbf{P}(\operatorname{im}(\mathbf{Q}^{\mathsf{T}})) = \mathbf{P}(\mathbb{R}^3) = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}.$$

This space has dimension 2. Hence

$$\operatorname{rk}(\mathbf{A}) = 2.$$

Therefore, the answer is $\boxed{2}$.

□

Problem 2015.1. Read the following passage and fill in the blank with the title of the mathematical classic, either in English or in Korean.

> For 2000 years, __________ was not only the core of mathematical education but also at the heart of Western culture. The most glowing tributes to __________ do not, in fact, come from mathematicians, but from philosophers, politicians, and others.
>
> Here are some examples:
>
> (a) He studied and nearly mastered the six books of __________ since he was a member of Congress. He regrets his want of education, and does what he can to supply the want.
>
> Abraham Lincoln, *Short Autobiography*
>
> (b) He studied __________ until he could demonstrate with ease all the propositions in the six books.
>
> *Herndon's Life of Lincoln*
>
> (c) At the age of eleven, I began __________. This was one of the great events of my life, as dazzling as first love. I had not imagined there was anything so delicious in the world.
>
> Bertrand Russell, *Autobiography*, Vol. 1

Solution. The passage is referring to Euclid's *Elements*, one of the most influential works in the history of mathematics.

For centuries, Euclid's *Elements* served as a standard model of rigorous mathematical reasoning and was central to mathematical education. The quotations from Abraham Lincoln and Bertrand Russell emphasize how deeply the book influenced not only mathematicians, but also philosophers, politicians, and intellectual culture more broadly.

Therefore, the answer is $\boxed{\textit{The Elements}\text{, or Euclid's } \textit{Elements}}$.

□

Problem 2015.2. Let $\mathbf{A}$ be an invertible square matrix. Suppose that, for a suitable permutation matrix $\mathbf{P}$, Gaussian elimination gives an LU-decomposition

$$\mathbf{PA} = \mathbf{LU}.$$

It may also happen that another permutation matrix $\widetilde{\mathbf{P}}$ gives another LU-decomposition

$$\widetilde{\mathbf{P}}\mathbf{A} = \widetilde{\mathbf{L}}\widetilde{\mathbf{U}}.$$

Show that the sets of absolute values of the pivots can be different. In other words, show by example that

$$\{|u_{kk}| : 1 \le k \le n\} \ne \{|\widetilde{u}_{kk}| : 1 \le k \le n\}.$$

Hint. Look for a counterexample in the 2×2 case.

Solution. Consider the invertible matrix

$$\mathbf{A} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{3} & 1 \end{pmatrix}.$$

With no row exchange, we have the LU-decomposition

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ \frac{1}{3} & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{5}{6} \end{pmatrix}.$$

Hence the absolute values of the pivots are

$$\left\{1, \frac{5}{6}\right\}.$$

Now let $\widetilde{\mathbf{P}}$ be the permutation matrix that swaps the two rows. Then

$$\widetilde{\mathbf{P}}\mathbf{A} = \begin{pmatrix} \frac{1}{3} & 1 \\ 1 & \frac{1}{2} \end{pmatrix}.$$

This matrix has the LU-decomposition

$$\widetilde{\mathbf{P}}\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 1 \\ 0 & -\frac{5}{2} \end{pmatrix}.$$

Hence the absolute values of the pivots are

$$\left\{\frac{1}{3}, \frac{5}{2}\right\}.$$

Therefore,

$$\left\{1, \frac{5}{6}\right\} \neq \left\{\frac{1}{3}, \frac{5}{2}\right\}.$$

This shows that the set of absolute values of the pivots can depend on the chosen row permutation.

$\square$

Problem 2015.3. In $\mathbb{R}^2$, the so-called Mercedes-Benz system is defined by

$$\{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2\}, \qquad \mathbf{v}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \end{pmatrix} \qquad (k = 0, 1, 2).$$

Find vectors $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2 \in \mathbb{R}^3$ such that they have the same length, are mutually orthogonal, have positive z-coordinates, and satisfy

$$\mathbf{v}_k = P_{xy}(\mathbf{w}_k) \qquad (k = 0, 1, 2),$$

where P_{xy} denotes the projection onto the xy-plane. Here $\mathbb{R}^2$ is regarded as the subspace of $\mathbb{R}^3$ with z-coordinate equal to 0.

Solution. Since $P_{xy}(\mathbf{w}_k) = \mathbf{v}_k$, each $\mathbf{w}_k$ must have the form

$$\mathbf{w}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \\ c \end{pmatrix}$$

for some positive constant c.

For $i \neq j$, the angle between $\mathbf{v}_i$ and $\mathbf{v}_j$ is $120°$, so

$$\mathbf{v}_i \cdot \mathbf{v}_j = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}.$$

Therefore,

$$\mathbf{w}_i \cdot \mathbf{w}_j = \mathbf{v}_i \cdot \mathbf{v}_j + c^2 = -\frac{1}{2} + c^2.$$

To make the vectors mutually orthogonal, we need

$$-\frac{1}{2} + c^2 = 0,$$

so

$$c = \frac{1}{\sqrt{2}}.$$

Hence one possible choice is

$$\boxed{\mathbf{w}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \\ \frac{1}{\sqrt{2}} \end{pmatrix} \qquad (k = 0, 1, 2)}.$$

These vectors all have squared length

$$1 + \frac{1}{2} = \frac{3}{2},$$

have positive z-coordinates, project to the given vectors $\mathbf{v}_k$, and are mutually orthogonal.

□

Problem 2015.4. This French mathematician was born in 1973 and received the Fields Medal in 2010 at the age of 36. On his website, he described himself as "Mathématicien, Professeur de l'Université de Lyon, Directeur de l'Institut Henri Poincaré." He is also famous for his stylish clothing.

Together with Clément Mouhot, he completed a theorem that played a central role in his Fields Medal-winning work. He later wrote a book about the process of developing this theorem, and the title of the book is also one of his nicknames.

Write the name of this mathematician, using the correct French spelling, and the title of the book.

Solution. The mathematician described in the problem is Cédric Villani.

Villani received the Fields Medal in 2010 for his work on nonlinear Landau damping and the Boltzmann equation. He is also widely recognized for his distinctive style, often including a cravat and a spider brooch.

The book mentioned in the problem is *Théorème vivant*, which may be translated as *Living Theorem*. It describes the process through which Villani and Clément Mouhot developed the theorem that contributed to Villani's Fields Medal-winning work.

Therefore, the answer is $\boxed{\text{Cédric Villani, \textit{Théorème vivant}}}$.

□

Problem 2015.5. On the front side of the Fields Medal, there is a profile of Archimedes together with the phrase by Marcus Manilius:

"TRANSIRE SUUM PECTUS MUNDOQUE POTIRI."

On the back side, there is another inscription together with the two solid figures that were engraved on Archimedes' tomb.

What are these two solid figures, and what is their volume ratio?

Solution. The two solid figures are a sphere and its circumscribed cylinder.

Suppose the sphere has radius R. Then the volume of the sphere is

$$V_S = \frac{4}{3}\pi R^3.$$

The circumscribed cylinder has radius R and height $2R$, so its volume is

$$V_C = \pi R^2 \cdot 2R = 2\pi R^3.$$

Therefore,

$$V_S : V_C = \frac{4}{3}\pi R^3 : 2\pi R^3 = 2 : 3.$$

Thus the two figures are a sphere and its circumscribed cylinder, and the volume ratio is $\boxed{2 : 3}$.

□

Problem 2015.6. Let $f : \mathbb{R} \to \mathbb{R}$ be a periodic function with period 2π. For $x \in (0, 2\pi]$, define

$$f(x) = \lceil \max(3, |\pi - x|) - 3 \rceil,$$

and extend f periodically, so that

$$f(x + 2\pi) = f(x)$$

for all $x \in \mathbb{R}$. Determine whether the following limit converges:

$$\lim_{n\to\infty} f(n).$$

Hint. Use $3.14 < \pi < 3.145$.

Solution. For $x \in (0, 2\pi]$, the function f takes only the values 0 and 1. More precisely,

$$f(x) = 1$$

when

$$0 < x < \pi - 3 \qquad \text{or} \qquad \pi + 3 < x \le 2\pi,$$

and

$$f(x) = 0$$

when

$$\pi - 3 \le x \le \pi + 3.$$

Thus f is a 2π-periodic function which is 1 near multiples of 2π, and 0 on the large middle interval.

Since 2π is irrational, the sequence of residues

$$n \pmod{2\pi}$$

is dense in $(0, 2\pi]$. Hence there are infinitely many positive integers n whose residues modulo 2π lie in

$$(0, \pi - 3) \cup (\pi + 3, 2\pi],$$

and for these integers we have

$$f(n) = 1.$$

There are also infinitely many positive integers n whose residues modulo 2π lie in

$$(\pi - 3, \pi + 3),$$

and for these integers we have

$$f(n) = 0.$$

Therefore, the sequence $f(n)$ has a subsequence equal to 1 and another subsequence equal to 0. Hence $f(n)$ does not converge as $n \to \infty$.

Thus the limit $\boxed{\text{does not converge}}$.

□

Problem 2015.7. This person is believed to have lived approximately from 624 BCE to 546 BCE. He is often regarded as one of the earliest figures worthy of being called a mathematician. He introduced the method of proof to geometric facts that had been known empirically through land surveying, helping to lay the foundations of geometry that were later systematized in Euclid's *Elements*.

The theorems attributed to him include:

(a) Vertical angles are equal.

(b) The base angles of an isosceles triangle are equal.

(c) Triangle congruence criteria such as side-angle-side and angle-side-angle.

(d) An angle inscribed in a semicircle is a right angle.

Who is this person?

Solution. The person described in the problem is Thales of Miletus.

Thales is traditionally regarded as one of the earliest Greek mathematicians. He is especially important because he is associated not merely with observing geometric facts, but with proving them. Several elementary geometric results are attributed to him, including the theorem that an angle inscribed in a semicircle is a right angle.

Therefore, the answer is $\boxed{\text{Thales of Miletus}}$.

□

Problem 2015.8. Which of the following mathematicians was born the earliest?

(a) Henri Poincaré
(b) Karl Weierstrass
(c) Julius Wilhelm Richard Dedekind
(d) Andrei Andreyevich Markov
(e) Georg Cantor

Solution. We compare their years of birth:

Karl Weierstrass	1815,
Julius Wilhelm Richard Dedekind	1831,
Georg Cantor	1845,
Henri Poincaré	1854,
Andrei Andreyevich Markov	1856.

Among these mathematicians, the one born earliest is Karl Weierstrass.

Therefore, the answer is $\boxed{\text{Karl Weierstrass}}$.

□

Problem 2015.9. The first International Congress of Mathematicians at which the Fields Medal was awarded was held in 1936. In which country was it held?

Solution. The Fields Medal was first awarded at the 1936 International Congress of Mathematicians. The 1936 ICM was held in Oslo, Norway.

Therefore, the answer is $\boxed{\text{Norway}}$.

□

Problem 2015.10. Find the sum of the distinct prime numbers appearing in the prime factorization of 15!.

Solution. The prime numbers appearing in the prime factorization of 15! are exactly the primes less than or equal to 15:

$$2, \quad 3, \quad 5, \quad 7, \quad 11, \quad 13.$$

Therefore, their sum is

$$2 + 3 + 5 + 7 + 11 + 13 = 41.$$

Thus the answer is $\boxed{41}$.

□

Problem 2015.11. In three-dimensional space, suppose that the lengths of the orthogonal projections of a line segment ℓ onto the yz-plane, the zx-plane, and the xy-plane are 5, 5, and 6, respectively. What is the length of ℓ?

Solution. Let the displacement vector of the line segment be

$$\mathbf{v} = (a, b, c).$$

Then the length of ℓ is

$$\|\mathbf{v}\| = \sqrt{a^2 + b^2 + c^2}.$$

The projection of $\mathbf{v}$ onto the yz-plane has length

$$\sqrt{b^2 + c^2} = 5.$$

Similarly, the projections onto the zx-plane and the xy-plane give

$$\sqrt{c^2 + a^2} = 5, \qquad \sqrt{a^2 + b^2} = 6.$$

Squaring and adding these three equations, we get

$$(b^2 + c^2) + (c^2 + a^2) + (a^2 + b^2) = 5^2 + 5^2 + 6^2.$$

Hence

$$2(a^2 + b^2 + c^2) = 25 + 25 + 36 = 86,$$

so

$$a^2 + b^2 + c^2 = 43.$$

Therefore, the length of ℓ is

$$\|\mathbf{v}\| = \sqrt{43}.$$

Thus the answer is $\boxed{\sqrt{43}}$.

□

Problem 2015.12. In three-dimensional space, consider the matrix obtained by rotating space by 30° around the line passing through the origin and the point $(1, 1, 2)$. How many real eigenvalues does this matrix have?

Solution. A rotation in $\mathbb{R}^3$ around an axis fixes every vector lying on the rotation axis. Therefore, the direction vector

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

is an eigenvector with eigenvalue 1.

On the plane perpendicular to the rotation axis, the matrix acts as a 30° rotation. A nontrivial rotation in a two-dimensional plane has no real eigenvalues unless the rotation angle is 0° or 180°. Since the angle here is 30°, the two remaining eigenvalues are non-real complex conjugates.

Hence the only real eigenvalue is 1. Therefore, the number of real eigenvalues is $\boxed{1}$.

□

Problem 2015.13. A right circular cone has base radius 3 and height 6. What is the maximum possible volume of a cylinder inscribed in this cone?

Solution. Let x be the radius of the inscribed cylinder, and let h be its height. By similarity of triangles in the cone, the radius of the horizontal cross-section decreases linearly from 3 to 0 as the height increases from 0 to 6. Hence

$$x = 3 - \frac{h}{2}.$$

Equivalently,

$$h = 6 - 2x.$$

Therefore, the volume of the cylinder is

$$V(x) = \pi x^2 h = \pi x^2(6 - 2x) = 6\pi x^2 - 2\pi x^3,$$

where $0 \le x \le 3$.

Differentiating, we get

$$V'(x) = 12\pi x - 6\pi x^2 = 6\pi x(2 - x).$$

Thus the critical point in the interval is

$$x = 2.$$

At this point,

$$h = 6 - 2 \cdot 2 = 2,$$

so the maximum volume is

$$V(2) = \pi \cdot 2^2 \cdot 2 = 8\pi.$$

Therefore, the maximum possible volume is $\boxed{8\pi}$.

□

Problem 2015.14. Suppose that a polynomial equation of degree 7 with real coefficients has three non-real complex roots whose absolute values are all different. How many real roots does the equation have?

Solution. For a polynomial with real coefficients, every non-real complex root appears together with its complex conjugate.

Suppose the three given non-real roots are

$$\alpha_1, \quad \alpha_2, \quad \alpha_3,$$

and their absolute values are all different. Then their complex conjugates

$$\overline{\alpha_1}, \quad \overline{\alpha_2}, \quad \overline{\alpha_3}$$

are also roots. Since

$$|\alpha_i| = |\overline{\alpha_i}|,$$

and the three given roots have distinct absolute values, these conjugate roots give three additional non-real roots.

Thus the polynomial has at least 6 non-real roots. Since the degree is 7, there is exactly one remaining root, and it must be real.

Therefore, the equation has $\boxed{1}$ real root.

□

Problem 2015.15. This mathematician was born in 1882 and died in 1935. When she was being considered for a professorship at the University of Göttingen, there was strong opposition. In response, Hilbert famously argued, "I do not see that the sex of the candidate is an argument against her admission as a Privatdozent. After all, we are a university, not a bathhouse."

She made remarkable contributions to several areas of mathematics, especially modern algebra, and much of twentieth-century abstract algebra was deeply influenced by her work. Who is this mathematician?

Solution. The mathematician described in the problem is Emmy Noether.

Noether made fundamental contributions to modern algebra, especially through her work on rings, ideals, and modules. She is also famous for Noether's theorem, which connects symmetries and conservation laws in physics.

The biographical details in the problem, including her connection with Göttingen and Hilbert's defense of her academic appointment, point to Emmy Noether.

Therefore, the answer is $\boxed{\text{Emmy Noether}}$.

□

Problem 2015.16. How many distinct terms appear when

$$(x_1 + x_2 + \cdots + x_7)^4$$

is expanded?

Solution. Each term in the expansion has the form

$$x_1^{k_1} x_2^{k_2} \cdots x_7^{k_7},$$

where

$$k_1 + k_2 + \cdots + k_7 = 4$$

and each k_i is a nonnegative integer.

Therefore, the number of distinct terms is the number of nonnegative integer solutions to

$$k_1 + k_2 + \cdots + k_7 = 4.$$

By the stars and bars formula, this number is

$$\binom{4+7-1}{7-1} = \binom{10}{6} = 210.$$

Thus the answer is $\boxed{210}$.

$\square$

Problem 2015.17. Let

$$\mathbf{v} = \mathbf{w} = (1, 2, 3, 4)$$

be row vectors. Consider the 4×4 matrix obtained as the product

$$\mathbf{v}^T\mathbf{w}.$$

Find the sum and the product of the eigenvalues of this matrix.

Solution. The matrix is

$$\mathbf{v}^T\mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \end{pmatrix}.$$

The sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to its trace. Therefore,

$$\sum \lambda_i = \operatorname{tr}(\mathbf{v}^T\mathbf{w}) = 1^2 + 2^2 + 3^2 + 4^2 = 30.$$

The product of the eigenvalues is equal to the determinant. Since $\mathbf{v}^T\mathbf{w}$ is an outer product of two vectors, it has rank 1. In particular, as a 4×4 matrix, it is not invertible, so

$$\det(\mathbf{v}^T\mathbf{w}) = 0.$$

Hence the product of the eigenvalues is

$$\prod \lambda_i = 0.$$

Therefore, the sum and product of the eigenvalues are $\boxed{30 \text{ and } 0}$, respectively.

$\square$

Problem 2015.18. Evaluate the infinite series

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!}.$$

Hint. Use

$$\frac{e^x - 1}{x} = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!},$$

then differentiate and substitute $x = 1$.

Solution. We use a simple telescoping identity:

$$\frac{n}{(n+1)!} = \frac{1}{n!} - \frac{1}{(n+1)!}.$$

Therefore,

$$\sum_{n=1}^{N} \frac{n}{(n+1)!} = \sum_{n=1}^{N} \left(\frac{1}{n!} - \frac{1}{(n+1)!} \right) = 1 - \frac{1}{(N+1)!}.$$

Taking $N \to \infty$, we obtain

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = 1.$$

Thus the answer is $\boxed{1}$.

□

Problem 2017.1. What are all possible positive integers that can occur as the number of faces of a regular polyhedron, that is, a Platonic solid?

Solution. There are exactly five Platonic solids:

Solid	Number of faces
Tetrahedron	4
Cube	6
Octahedron	8
Dodecahedron	12
Icosahedron	20

Therefore, the possible numbers of faces of a regular polyhedron are

$$\boxed{4,\ 6,\ 8,\ 12,\ 20}.$$

□

Problem 2017.2. Let F_n be the n-th Fibonacci number. What is

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n}?$$

Solution. The Fibonacci numbers satisfy

$$F_{n+1} = F_n + F_{n-1}.$$

Suppose

$$L = \lim_{n\to\infty} \frac{F_{n+1}}{F_n}.$$

Then

$$\frac{F_{n+1}}{F_n} = 1 + \frac{F_{n-1}}{F_n}.$$

Taking limits gives

$$L = 1 + \frac{1}{L}.$$

Hence

$$L^2 = L + 1,$$

so

$$L = \frac{1 \pm \sqrt{5}}{2}.$$

Since the Fibonacci numbers are positive for large n, the limit must be positive. Therefore,

$$L = \frac{1+\sqrt{5}}{2}.$$

Thus the answer is $\boxed{\dfrac{1+\sqrt{5}}{2}}$.

$\square$

Problem 2017.3. For a positive integer n, define the $n \times n$ matrix $\mathbf{C}$ by

$$C_{ij} = \binom{i+j-2}{i-1} \qquad (1 \le i, j \le n).$$

Now define another $n \times n$ matrix $\mathbf{B}$ by

$$B_{ij} = \begin{cases} C_{ij}, & (i,j) \ne (n,n), \\ C_{ij} + 2016, & (i,j) = (n,n). \end{cases}$$

Find $\det \mathbf{B}$.

Solution. We first compute the determinant of $\mathbf{C}$. By Vandermonde's identity,

$$\binom{i+j-2}{i-1} = \sum_{k=1}^{\min(i,j)} \binom{i-1}{k-1}\binom{j-1}{k-1}.$$

Hence

$$\mathbf{C} = \mathbf{L}\mathbf{L}^T,$$

where

$$L_{ik} = \begin{cases} \binom{i-1}{k-1}, & k \le i, \\ 0, & k > i. \end{cases}$$

The matrix $\mathbf{L}$ is lower triangular and all of its diagonal entries are 1. Therefore,

$$\det \mathbf{L} = 1,$$

and so

$$\det \mathbf{C} = \det(\mathbf{L}\mathbf{L}^T) = (\det \mathbf{L})^2 = 1.$$

The matrix $\mathbf{B}$ differs from $\mathbf{C}$ only in the (n,n)-entry. Since the determinant is linear in each entry, we have

$$\det \mathbf{B} = \det \mathbf{C} + 2016 \cdot M_{nn},$$

where M_{nn} is the (n,n)-minor of $\mathbf{C}$.

This minor is the determinant of the upper-left $(n-1) \times (n-1)$ submatrix of $\mathbf{C}$, which has the same form as $\mathbf{C}$ with size $n-1$. Therefore,

$$M_{nn} = 1.$$

Hence

$$\det \mathbf{B} = 1 + 2016 \cdot 1 = 2017.$$

Thus the answer is $\boxed{2017}$.

$\square$

Problem 2017.4. In 2017, this prize was awarded to the French mathematician Yves Meyer. The Norwegian Academy of Science and Letters announced on March 21 that Meyer would receive the prize, worth six million Norwegian kroner, in recognition of his decisive contributions to the development of wavelet theory, which is used in areas ranging from gravitational wave detection to movie file compression.

Often called the Nobel Prize of mathematics, what is this prize?

Solution. The prize described in the problem is the Abel Prize.

The Abel Prize is awarded by the Norwegian Academy of Science and Letters and is often described as one of the most prestigious prizes in mathematics. In 2017, it was awarded to Yves Meyer for his fundamental contributions to the mathematical theory of wavelets.

Therefore, the answer is $\boxed{\text{Abel Prize}}$.

□

Problem 2017.5. Express the integral

$$\int_0^1 x^m(1-x)^n\,dx$$

in terms of a binomial coefficient closely related to the positive integers m and n.

Solution. This integral is a beta integral. For positive integers m and n, we have

$$\int_0^1 x^m(1-x)^n\,dx = \frac{m!n!}{(m+n+1)!}.$$

This is the factorial form of the answer.

To express it using a binomial coefficient, recall that

$$\binom{m+n}{m} = \frac{(m+n)!}{m!n!}.$$

Hence

$$\frac{m!n!}{(m+n+1)!} = \frac{1}{m+n+1}\cdot\frac{m!n!}{(m+n)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Therefore,

$$\int_0^1 x^m(1-x)^n\,dx = \frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Thus the answer is $\boxed{\frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}}$.

□

Problem 2018.1. The following people are this year's Fields Medalists. Excluding choice (5), who is the second oldest among them?

(1) Caucher Birkar
(2) Alessio Figalli
(3) Peter Scholze
(4) Akshay Venkatesh
(5) The person who stole Birkar's medal

Solution. The 2018 Fields Medalists were

(1)	Caucher Birkar	born in 1978,
(2)	Alessio Figalli	born in 1984,
(3)	Peter Scholze	born in 1987,
(4)	Akshay Venkatesh	born in 1981.

Among these four, the second oldest is Akshay Venkatesh. Therefore, the answer is

$$\boxed{(4)}.$$

Choice (5) is a joke referring to the incident in which Caucher Birkar's Fields Medal was stolen shortly after the 2018 Fields Medal ceremony. Birkar later received a replacement medal. □

Problem 2018.2. According to William Thurston's geometrization conjecture, a 3-dimensional manifold can be cut along spheres and tori so that each resulting piece has one of n possible geometric structures. What is n?

Solution. The 3-dimensional geometric structures appearing in Thurston's geometrization conjecture are the eight Thurston geometries:

$$S^3, \quad \mathbb{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}_2(\mathbb{R})}, \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Hence $n = 8$, so the answer is

$$\boxed{8}.$$

□

Problem 2018.3. Find the sum of the following infinite series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \cdots.$$

Solution. The given series is one half of the alternating harmonic series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right).$$

Recall the Taylor expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots.$$

Substituting $x = 1$, we obtain

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

Therefore,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{\ln 2}{2}.$$

Thus the answer is $\boxed{\frac{\ln 2}{2}}$.

□

Problem 2018.4. Among the following five Fields Medalists, which one does not have a Korean doctoral student?

(1) William Thurston
(2) Maxim Kontsevich
(3) Grigory Margulis
(4) Terence Tao
(5) Efim Zelmanov

Solution. The intended answer is Maxim Kontsevich.

This problem is not asking for a theorem but for advisor–student data about Fields Medalists and Korean mathematicians. In a Science Quiz setting, it is best treated as a memorization-style question: among the five listed Fields Medalists, the exceptional name in the intended answer key is

$$\boxed{\text{(2) Maxim Kontsevich}}.$$

□

Problem 2019.1. Consider the 2019×2019 matrix whose diagonal entries are all 0 and whose off-diagonal entries are all 1. Find all eigenvalues of this matrix.

Solution. Let $\mathbf{A}$ be the given matrix, and let $\mathbf{J}$ be the 2019×2019 matrix whose entries are all 1. Then

$$\mathbf{A} = \mathbf{J} - \mathbf{I}.$$

First, consider the vector

$$\mathbf{1} = (1, 1, \ldots, 1)^T.$$

Since

$$\mathbf{J}\mathbf{1} = 2019\mathbf{1},$$

we get

$$\mathbf{A}\mathbf{1} = (\mathbf{J} - \mathbf{I})\mathbf{1} = 2019\mathbf{1} - \mathbf{1} = 2018\mathbf{1}.$$

Hence 2018 is an eigenvalue.

Now take any vector $\mathbf{v} = (v_1, \ldots, v_{2019})^T$ satisfying

$$v_1 + \cdots + v_{2019} = 0.$$

Then

$$\mathbf{J}\mathbf{v} = \mathbf{0},$$

and therefore

$$\mathbf{A}\mathbf{v} = (\mathbf{J} - \mathbf{I})\mathbf{v} = -\mathbf{v}.$$

Thus -1 is an eigenvalue. The subspace of vectors whose coordinates sum to 0 has dimension 2018, so the multiplicity of -1 is 2018.

Therefore, the eigenvalues are

$$\boxed{2018 \text{ with multiplicity } 1, \qquad -1 \text{ with multiplicity } 2018}.$$

□

Problem 2019.2. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2}\, dx.$$

Solution. Let

$$u = 2x.$$

Then

$$dx = \frac{1}{2}\, du.$$

Therefore,

$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2}\, dx = \int_{-\infty}^{\infty} \frac{8}{1+u^2} \cdot \frac{1}{2}\, du = 4 \int_{-\infty}^{\infty} \frac{1}{1+u^2}\, du.$$

Since
$$\int_{-\infty}^{\infty} \frac{1}{1+u^2}\, du = \pi,$$
we obtain
$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2}\, dx = 4\pi.$$

Thus the answer is $\boxed{4\pi}$.

□

Problem 2019.3. Compute the value of $\pi + e$ up to five decimal places.

Solution. Using the approximations
$$\pi \approx 3.14159265, \qquad e \approx 2.71828183,$$
we get
$$\pi + e \approx 3.14159265 + 2.71828183 = 5.85987448.$$
Therefore, up to five decimal places,
$$\pi + e \approx 5.85987.$$

Thus the answer is $\boxed{5.85987}$.

□

Problem 2020.1. The International Congress of Mathematicians, abbreviated as ICM, is a congress for mathematicians from around the world. It is organized every four years by the International Mathematical Union and may be thought of as a kind of Olympic Games for mathematicians. The 2014 ICM was held in South Korea, and the 2018 ICM was held in Brazil. As of 2019, in which country was the 2022 ICM scheduled to be held?

Solution. ICM stands for the International Congress of Mathematicians. It is the main international congress in mathematics and is organized every four years by the International Mathematical Union.

The 2014 ICM was held in Seoul, South Korea, and the 2018 ICM was held in Rio de Janeiro, Brazil. At the time this problem was asked in 2019, the 2022 ICM was scheduled to be held in Saint Petersburg, Russia. Therefore, the answer is $\boxed{\text{Russia}}$.

In reality, after Russia's invasion of Ukraine in 2022, the 2022 ICM was not held in Saint Petersburg. It was eventually held as a virtual event, with the award ceremony taking place in Helsinki, Finland. Thus, the answer to this problem should be understood according to the planned host country at the time of the 2019 competition.

□

Problem 2020.2. A fair six-sided die has the numbers $-2, 0, 2, 3, 4, 5$ written on its faces. The die is rolled 10 times, and X is defined to be the product of the 10 numbers obtained. What is the expected value of X?

Solution. Let Y be the number obtained from one roll of the die. Since all six faces are equally likely,
$$\mathbb{E}[Y] = \frac{-2+0+2+3+4+5}{6} = 2.$$
Now let $Y_1, Y_2, \ldots, Y_{10}$ be the numbers obtained from the 10 rolls. Then
$$X = Y_1 Y_2 \cdots Y_{10}.$$

Since the rolls are independent, the random variables $Y_1, Y_2, \ldots, Y_{10}$ are independent. Therefore,

$$\mathbb{E}[X] = \mathbb{E}[Y_1 Y_2 \cdots Y_{10}] = \mathbb{E}[Y_1]\mathbb{E}[Y_2] \cdots \mathbb{E}[Y_{10}].$$

All the Y_i have the same distribution as Y, so

$$\mathbb{E}[X] = (\mathbb{E}[Y])^{10} = 2^{10} = 1024.$$

Thus the answer is $\boxed{1024}$.

□

Problem 2020.3. Which of the following Fields Medalists was not serving as a professor at the time?

(1) Cédric Villani
(2) Elon Lindenstrauss
(3) Peter Scholze
(4) Martin Hairer
(5) Terence Tao

Solution. At the time this problem was asked, Cédric Villani was active as a politician in France rather than serving as a professor.

The other choices were mathematicians holding professorial positions. Therefore, the answer is $\boxed{\text{(1) Cédric Villani}}$.

□

Problem 2020.4. In a three-dimensional gravitational field, when an object moves from a point $\mathbf{a}$ to another point $\mathbf{b}$, the change in its potential energy is independent of the path taken from $\mathbf{a}$ to $\mathbf{b}$. Which of the following mathematical facts is least directly related to this physical phenomenon?

(1) A curl-free vector field is the gradient of some potential function.
(2) Stokes' theorem
(3) The Laplacian of $\dfrac{1}{|\mathbf{x}|}$ is a delta function, up to a constant factor.
(4) The curl of a gradient is identically zero.

Solution. The statement that the change in potential energy is independent of the path means that the gravitational field is a conservative vector field.

Let $\mathbf{F}$ be the force field. The change in potential energy from $\mathbf{a}$ to $\mathbf{b}$ is given by

$$\Delta U = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot \mathrm{d}\mathbf{r}.$$

For this quantity to be independent of the path, the force field should be expressible as the negative gradient of a potential function:

$$\mathbf{F} = -\nabla U.$$

Thus choice (1) is directly related to the phenomenon.

Also, the curl of a gradient is always zero:

$$\nabla \times \nabla U = \mathbf{0}.$$

Hence choice (4) is also directly related. Stokes' theorem says that

$$\oint_{\partial S} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, \mathrm{d}S.$$

Therefore, if $\nabla \times \mathbf{F} = \mathbf{0}$, then the line integral around a closed curve is zero, which is another way to understand path independence. Thus choice (2) is also related.

On the other hand,

$$\Delta\left(\frac{1}{|\mathbf{x}|}\right) = -4\pi\delta(\mathbf{x})$$

is related to the fundamental solution of the Laplacian and to Poisson's equation for gravitational potential. Although it is related to gravity, it is not the main mathematical reason why the potential energy change is independent of path. Therefore, the answer is $\boxed{(3)}$.

□

Problem 2020.5. Among the following mathematics-related academic institutions, which one is actually a degree-granting school?

(1) Institut des Hautes Études Scientifiques, IHES
(2) École normale supérieure
(3) Mathematical Sciences Research Institute, MSRI
(4) Institute for Advanced Study, IAS
(5) Collège de France

Solution. École normale supérieure is one of France's most prestigious higher education institutions. It operates educational programs and is naturally regarded as a degree-granting school.

By contrast, IHES, MSRI, and IAS are research institutes rather than degree-granting schools. Collège de France is also a famous academic institution that offers public lectures, but it does not grant degrees.

Therefore, the answer is $\boxed{\text{(2) École normale supérieure}}$.

□

Problem 2021.1. Which is larger, A or B, where

$$A = 1 + \frac{1}{1 + \frac{1}{1 - \frac{1}{1+4}}}, \qquad B = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1+3}}}?$$

Solution. First, we compute A:

$$A = 1 + \frac{1}{1 + \frac{1}{1 - \frac{1}{5}}} = 1 + \frac{1}{1 + \frac{1}{\frac{4}{5}}} = 1 + \frac{1}{1 + \frac{5}{4}} = 1 + \frac{1}{\frac{9}{4}} = 1 + \frac{4}{9} = \frac{13}{9}.$$

Next, we compute B:

$$B = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{4}}} = 1 + \frac{1}{1 + \frac{1}{\frac{5}{4}}} = 1 + \frac{1}{1 + \frac{4}{5}} = 1 + \frac{1}{\frac{9}{5}} = 1 + \frac{5}{9} = \frac{14}{9}.$$

Since

$$\frac{13}{9} < \frac{14}{9},$$

we have $A < B$. Therefore, the answer is $\boxed{B}$.

□

Problem 2021.2. Let

$$A(n) = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^{n+1}.$$

What is the integer closest to $A(7)$?

Solution. Recall Binet's formula for the Fibonacci sequence:

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$

Here

$$A(7) = \frac{\varphi^8}{\sqrt{5}}.$$

By Binet's formula,

$$F_8 = \frac{\varphi^8 - \psi^8}{\sqrt{5}}.$$

Since $|\psi| < 1$, the term $\dfrac{\psi^8}{\sqrt{5}}$ is very small. Thus

$$A(7) = \frac{\varphi^8}{\sqrt{5}} = F_8 + \frac{\psi^8}{\sqrt{5}}$$

is very close to F_8.

The Fibonacci numbers are

$$F_0 = 0, \quad F_1 = 1, \quad F_2 = 1, \quad F_3 = 2, \quad F_4 = 3, \quad F_5 = 5, \quad F_6 = 8, \quad F_7 = 13, \quad F_8 = 21.$$

Therefore, the integer closest to $A(7)$ is $\boxed{21}$.

□

Problem 2021.3. Choose all groups among the following that are not finitely generated:

(1) The fundamental group of an open unit disk with finitely many punctures.
(2) $\mathrm{SL}_2(\mathbb{Z})$, the group of 2×2 integer matrices with determinant 1.
(3) $\mathbb{C}^\times$, the group of nonzero complex numbers under multiplication.
(4) The Monster group, the largest sporadic finite simple group.

Solution. First, consider an open unit disk with finitely many punctures. If the removed points are $p_1, \dots, p_n$, then the fundamental group is isomorphic to the free group on n generators:

$$\pi_1\left(D \setminus \{p_1, \dots, p_n\}\right) \cong F_n.$$

Hence the group in choice (1) is finitely generated.

The group $\mathrm{SL}_2(\mathbb{Z})$ is also finitely generated. For example, it is generated by the two matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Thus choice (2) is finitely generated.

On the other hand, $\mathbb{C}^\times$ is an uncountable abelian group. Every finitely generated group is countable, so $\mathbb{C}^\times$ cannot be finitely generated. Therefore, choice (3) is not finitely generated.

Finally, the Monster group is a finite group. Every finite group is finitely generated, since one may simply take all of its elements as generators. Hence choice (4) is finitely generated.

Therefore, the only group in the list that is not finitely generated is $\boxed{(3)\ \mathbb{C}^\times}$.

□

Problem 2021.4. Let
$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$
Find all eigenvalues of $\mathbf{A}$ together with their multiplicities. Also determine whether $\mathbf{A}$ is diagonalizable.

Solution. We can write $\mathbf{A}$ as
$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}.$$
Thus $\mathbf{A}$ is a rank-one matrix.

First, let
$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$
Then
$$\mathbf{A}\mathbf{v} = \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix} = 6\mathbf{v}.$$
Hence 6 is an eigenvalue of $\mathbf{A}$.

Since $\mathbf{A}$ has rank 1, its nullity is 2. Therefore, the eigenspace corresponding to the eigenvalue 0 has dimension 2. Indeed,
$$\mathbf{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+2y+3z \\ x+2y+3z \\ x+2y+3z \end{pmatrix},$$
so
$$E_0 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x+2y+3z = 0 \right\}.$$
Thus the geometric multiplicity of 0 is 2.

Now
$$\operatorname{tr}(\mathbf{A}) = 1+2+3 = 6.$$
Since the sum of the eigenvalues, counted with algebraic multiplicity, is equal to the trace, the eigenvalues are
$$6, \quad 0, \quad 0.$$
Hence the algebraic multiplicities are
$$6 : 1, \qquad 0 : 2.$$

Finally, the eigenspace for 6 has dimension 1, and the eigenspace for 0 has dimension 2. The total dimension of the eigenspaces is therefore
$$1 + 2 = 3.$$
Hence there exists a basis of $\mathbb{R}^3$ consisting of eigenvectors of $\mathbf{A}$. Therefore, $\mathbf{A}$ is diagonalizable.

Thus the eigenvalues are $\boxed{6, 0, 0}$, with algebraic multiplicities 1 and 2 respectively, and $\boxed{\mathbf{A} \text{ is diagonalizable}}$.

□

Problem 2022.1. Which of the following numbers is the largest?

(1) e^{π}
(2) π^{e}
(3) 7π
(4) $\binom{7}{2}$
(5) $4!$

Solution. We compare the values directly:

$$e^{\pi} \approx 23.1407, \qquad \pi^{e} \approx 22.4592, \qquad 7\pi \approx 21.9911.$$

Also,

$$\binom{7}{2} = \frac{7 \cdot 6}{2} = 21, \qquad 4! = 24.$$

Therefore, the largest number among the choices is $\boxed{(5)\ 4!}$.

□

Problem 2022.2. Three people store a jointly owned treasure in a safe and want to lock it. They want no single person to be able to open the safe alone, but any two or more people should always be able to open it together. This can be done by putting n locks on the safe, making m copies of the key for each lock, and distributing the keys appropriately among the three people. Find n and m so that the number of locks n is minimized.

Solution. Let the three people be A, B, and C.

Since any two people should be able to open the safe, the key to each lock must be given to at least two people. If a certain lock had its key given to only one person, then the other two people would not be able to open that lock together.

Thus, for each lock, exactly one of the three people should not receive the key. On the other hand, no single person should be able to open the safe alone. Therefore, each person must be missing the key to at least one lock.

Hence we need at least three locks: one lock whose key is not given to A, one lock whose key is not given to B, and one lock whose key is not given to C. This lower bound is achievable by distributing the keys as follows:

	A	B	C
L_A	×	✓	✓
L_B	✓	×	✓
L_C	✓	✓	×

That is, the key to L_A is given to B and C, the key to L_B is given to A and C, and the key to L_C is given to A and B.

Then each person is unable to open one of the locks, so no one can open the safe alone. However, any two people together can open all three locks. Therefore, the minimum is achieved by $\boxed{n = 3,\ m = 2}$.

□

Problem 2022.3. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2 + 7}\, \mathrm{d}x.$$

Solution. Let

$$u = x - 2.$$

Then the limits of integration remain $-\infty$ and ∞, so

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2+7}\,\mathrm{d}x = 3\int_{-\infty}^{\infty} \frac{1}{u^2+7}\,\mathrm{d}u.$$

We use the standard formula

$$\int_{-\infty}^{\infty} \frac{1}{u^2+a^2}\,\mathrm{d}u = \frac{\pi}{a}.$$

Here $a=\sqrt{7}$, and therefore

$$3\int_{-\infty}^{\infty} \frac{1}{u^2+7}\,\mathrm{d}u = 3\cdot\frac{\pi}{\sqrt{7}} = \frac{3\pi}{\sqrt{7}}.$$

Thus the answer is $\boxed{\dfrac{3\pi}{\sqrt{7}}}$.

□

Problem 2022.4. A couple has two children.

(a) Given that one of the children is a daughter, what is the probability that the other child is also a daughter?

(b) Given that one of the children is a daughter named Youngmi, what is the probability that the other child is also a daughter?

Solution. Let G denote a girl and B denote a boy. If we distinguish the two children by birth order, then the possible gender combinations are

$$GG, \quad GB, \quad BG, \quad BB.$$

For part (a), the condition that one child is a daughter means that at least one child is a girl. Thus the possible cases are

$$GG, \quad GB, \quad BG.$$

Among these three cases, only GG has the property that the other child is also a daughter. Therefore,

$$\mathbb{P}(\text{both are daughters} \mid \text{at least one is a daughter}) = \frac{1}{3}.$$

For part (b), the information that one child is a daughter named Youngmi is more specific. In the intended interpretation of the problem, this information identifies one particular child. Then the gender of the other child is still equally likely to be G or B. Hence

$$\mathbb{P}(\text{the other child is a daughter}) = \frac{1}{2}.$$

Therefore, the answers are $\boxed{\text{(a) } \frac{1}{3},\ \text{(b) } \frac{1}{2}}$.

□

Problem 2022.5. Which of the following conjectures was not solved by Professor June Huh?

(1) Read's conjecture
(2) Brylawski's conjecture
(3) Poincaré conjecture
(4) Rota's conjecture
(5) Okounkov's conjecture

Solution. Professor June Huh is known for connecting combinatorics with algebraic geometry and using these ideas to solve several major conjectures in combinatorics. His work is related to conjectures such as Read's conjecture, Brylawski's conjecture, Rota's conjecture, and Okounkov's conjecture.

On the other hand, the Poincaré conjecture is a famous problem in 3-dimensional topology. It was proved by Grigori Perelman using Ricci flow, not by June Huh.

Therefore, the conjecture in the list that was not solved by June Huh is

$$\boxed{\text{(3) Poincaré conjecture}}.$$

□

Problem 2022.6. This mathematician, often called the father of linear programming, is famous for solving difficult problems while he was a doctoral student. One day, while studying at UC Berkeley, he arrived late to a class. The statistics professor Jerzy Neyman had written two famous unsolved problems on the blackboard, but the student thought they were homework problems. He took them home, solved them, and submitted complete solutions a few days later, thinking only that he had turned in the homework late. Who was this mathematician?

Solution. The mathematician described in the problem is George Dantzig.

George Dantzig is known as one of the founders of linear programming and as the creator of the simplex method. For this reason, he is often called the father of linear programming.

The anecdote in the problem is also a famous story about Dantzig. While he was a doctoral student at UC Berkeley, he arrived late to Jerzy Neyman's statistics class and copied down two problems written on the blackboard, believing that they were homework problems. He later submitted solutions, not realizing that the problems were actually open problems in statistics.

Therefore, the answer is $\boxed{\text{George Dantzig}}$.

□

Problem 2022.7. Two different coins are each tossed three times. Let X and Y be the numbers of heads obtained from the two coins, respectively. What is the probability that $X = Y$?

Solution. For each coin, the number of heads obtained in three tosses follows the binomial distribution

$$X, Y \sim \mathrm{Bin}\left(3, \frac{1}{2}\right).$$

Also, X and Y are independent. Therefore,

$$\mathbb{P}(X = Y) = \sum_{k=0}^{3} \mathbb{P}(X = k)\mathbb{P}(Y = k).$$

For each $k = 0, 1, 2, 3$, we have

$$\mathbb{P}(X = k) = \binom{3}{k}\left(\frac{1}{2}\right)^3,$$

and the same formula holds for Y. Hence

$$\mathbb{P}(X = Y) = \sum_{k=0}^{3} \left(\binom{3}{k}\left(\frac{1}{2}\right)^3\right)^2 = \frac{1}{64}\sum_{k=0}^{3}\binom{3}{k}^2.$$

Thus

$$\mathbb{P}(X = Y) = \frac{1}{64}\left(\binom{3}{0}^2 + \binom{3}{1}^2 + \binom{3}{2}^2 + \binom{3}{3}^2\right) = \frac{1+9+9+1}{64} = \frac{20}{64} = \frac{5}{16}.$$

Therefore, the answer is $\boxed{\dfrac{5}{16}}$.

$\square$

Problem 2023.1. In which year was the Fields Medal, won by Professor June Huh, first awarded?

Solution. The Fields Medal is one of the most prestigious awards in mathematics, and it is awarded by the International Mathematical Union. Professor June Huh received the Fields Medal in 2022.

The Fields Medal was first awarded in 1936. The first two recipients were Lars Ahlfors and Jesse Douglas.

Therefore, the answer is $\boxed{1936}$.

$\square$

Problem 2023.2. Consider the 6×6 square grid with vertices $(0,0)$, $(6,0)$, $(0,6)$, and $(6,6)$. A path starts at $(0,0)$ and ends at $(6,6)$, moving only one unit to the right or one unit upward at each step. How many such paths never go above the line $x = y$? Equivalently, how many such paths stay in the region

$$\{(x,y) \in \mathbb{R}^2 : y \leq x\}?$$

Solution. To go from $(0,0)$ to $(6,6)$, a path must consist of 6 right steps and 6 upward steps. Without the condition $y \leq x$, the total number of such paths is

$$\binom{12}{6}.$$

We now count the paths that do go above the line $y = x$. By the reflection principle, the number of paths from $(0,0)$ to $(6,6)$ that at some point satisfy $y > x$ is

$$\binom{12}{5}.$$

Therefore, the number of paths that always stay in the region $y \leq x$ is

$$\binom{12}{6} - \binom{12}{5}.$$

Computing this gives

$$\binom{12}{6} - \binom{12}{5} = 924 - 792 = 132.$$

Thus the answer is $\boxed{132}$.

$\square$

Problem 2023.3. A point on the surface of the Earth has the following property: starting from that point, one moves 5km south, then 5km east, and then 5km north, and returns to the original point. How many such points are there on the surface of the Earth?

Assume that the surface of the Earth is a perfect two-dimensional sphere, and that the North Pole and the South Pole are antipodal points. At the North Pole, the directions north, east, and west are not defined, and at the South Pole, the directions south, east, and west are not defined. Starting points for which one of these undefined directions occurs during the movement are excluded.

Solution. The North Pole is one such point. If we start at the North Pole, move 5km south, then move 5km east along a latitude circle, and finally move 5km north, we return to the North Pole.

There are also many such points near the South Pole. Let Q be the point reached after moving 5km south from the starting point. Suppose that the latitude circle passing through Q has circumference

$$\frac{5}{n}\text{km}$$

for some positive integer n. Then moving 5km east from Q goes exactly n full times around that latitude circle and returns to Q. After that, moving 5km north returns to the original starting point.

Near the South Pole, the circumferences of latitude circles approach 0. Hence, for every positive integer n, there is a latitude circle near the South Pole whose circumference is exactly $\frac{5}{n}$km. For each such latitude circle, every point lying 5km north of it gives a valid starting point.

Therefore, there are not just one but infinitely many such points on the Earth's surface. In fact, this construction gives entire latitude circles of valid starting points, so there are uncountably many such points. For the usual quiz answer, the essential conclusion is

$$\boxed{\text{infinitely many}}.$$

□

Problem 2023.4. Let $f : (-1, 1) \to \mathbb{R}$ be a continuous function. For each $\lambda > 0$, define

$$f_\lambda(x) := \lambda f\left(\frac{x}{\lambda}\right),$$

where f_λ is defined for $x \in (-\lambda, \lambda)$.

Suppose that for every sequence $\lambda_i \to \infty$, the functions f_{λ_i} converge locally uniformly to 0 on $\mathbb{R}$. Determine whether the following statement is true or false:

$$f \text{ is differentiable at } 0 \text{ and } f'(0) = 0.$$

Solution. We prove that the statement is true.

First, evaluate at $x = 0$. For every sequence $\lambda_i \to \infty$, we have

$$f_{\lambda_i}(0) = \lambda_i f(0).$$

Since f_{λ_i} converges locally uniformly to 0, in particular it converges pointwise to 0 at $x = 0$. Hence

$$\lambda_i f(0) \to 0.$$

This is possible for all sequences $\lambda_i \to \infty$ only if

$$f(0) = 0.$$

Now we show that $f'(0) = 0$. Let $h_i \to 0$ be any sequence with $h_i \neq 0$. For all sufficiently large i, we have $h_i \in (-1, 1)$. Define

$$\lambda_i = \frac{1}{|h_i|}.$$

Then $\lambda_i \to \infty$, and

$$h_i = \frac{\operatorname{sgn}(h_i)}{\lambda_i}.$$

By the assumption, $f_{\lambda_i} \to 0$ locally uniformly on $\mathbb{R}$. In particular, this convergence is uniform on the compact set $\{-1, 1\}$. Therefore,

$$f_{\lambda_i}(\operatorname{sgn}(h_i)) \to 0.$$

But

$$f_{\lambda_i}(\operatorname{sgn}(h_i)) = \lambda_i f\left(\frac{\operatorname{sgn}(h_i)}{\lambda_i}\right) = \lambda_i f(h_i).$$

Hence

$$\lambda_i f(h_i) \to 0.$$

Since $f(0) = 0$, we get

$$\left|\frac{f(h_i) - f(0)}{h_i}\right| = \left|\frac{f(h_i)}{h_i}\right| = \lambda_i |f(h_i)| \to 0.$$

Thus

$$\frac{f(h_i) - f(0)}{h_i} \to 0$$

for every sequence $h_i \to 0$ with $h_i \neq 0$. Therefore, f is differentiable at 0 and

$$f'(0) = 0.$$

Thus the answer is $\boxed{\text{True}}$.

□

Problem 2024.1. When all natural numbers from 100 to 999 are written in a row, how many times does the digit 0 appear?

Solution. Every natural number from 100 to 999 is a three-digit number.

In a three-digit number, the hundreds digit cannot be 0. Therefore, the digit 0 can appear only in the tens digit or the units digit.

First, count the numbers whose tens digit is 0. The hundreds digit can be one of $1, 2, \ldots, 9$, and the units digit can be one of $0, 1, \ldots, 9$. Hence there are

$$9 \cdot 10 = 90$$

such numbers.

Next, count the numbers whose units digit is 0. The hundreds digit can be one of $1, 2, \ldots, 9$, and the tens digit can be one of $0, 1, \ldots, 9$. Hence there are again

$$9 \cdot 10 = 90$$

such numbers.

Therefore, the total number of appearances of the digit 0 is

$$90 + 90 = 180.$$

Thus the answer is $\boxed{180}$.

□

Problem 2024.2. Which of the following people was born the latest?

(1) John von Neumann
(2) Alan Turing
(3) John Nash
(4) Pierre de Fermat

Solution. We compare their years of birth:

(1) John von Neumann: born in 1903,
(2) Alan Turing: born in 1912,
(3) John Nash: born in 1928,
(4) Pierre de Fermat: born in 1607.

Among these four, the person born the latest is John Nash.

Therefore, the answer is $\boxed{\text{(3) John Nash}}$.

□

Problem 2024.4. Let C be a convex subset of $\mathbb{R}^3$. Here convex means that for any two points $\mathbf{p}, \mathbf{q} \in C$, the line segment joining $\mathbf{p}$ and $\mathbf{q}$ is also contained in C.

Suppose that the following two facts are known:

(i) The intersection of C with the xy-plane is

$$C \cap \{z = 0\} = \{(x, y, 0) : x^2 + 4y^2 \leq 1\}.$$

(ii) The set C contains the z-axis.

Can the volume of C in the region $-1 \leq z \leq 1$ be determined?

(1) This cannot be determined.
(2) π
(3) 2π
(4) 4π

Solution. Let

$$E = \{(x, y) \in \mathbb{R}^2 : x^2 + 4y^2 \leq 1\}.$$

By condition (i),

$$C \cap \{z = 0\} = E \times \{0\}.$$

The ellipse E has semiaxes 1 and $\dfrac{1}{2}$, so its area is

$$\text{area}(E) = \pi \cdot 1 \cdot \frac{1}{2} = \frac{\pi}{2}.$$

For each height z, define the horizontal cross-section

$$C_z = \{(x, y) \in \mathbb{R}^2 : (x, y, z) \in C\}.$$

We claim that C_z has the same area as E for every z.

First, we show that $C_z \subseteq E$. Suppose that $(x, y, z) \in C$ and

$$x^2 + 4y^2 > 1.$$

Since C contains the z-axis, the point $(0, 0, t)$ belongs to C for every $t \in \mathbb{R}$. By convexity, the line segment joining (x, y, z) and $(0, 0, t)$ is contained in C.

For any $\alpha \in (0, 1)$, we may choose

$$t = -\frac{\alpha z}{1 - \alpha}$$

so that this segment intersects the xy-plane at the point

$$(\alpha x, \alpha y, 0).$$

Since α can be chosen sufficiently close to 1, we may arrange that

$$(\alpha x)^2 + 4(\alpha y)^2 > 1.$$

This would give a point of $C \cap \{z = 0\}$ lying outside $E \times \{0\}$, contradicting condition (i). Therefore,

$$C_z \subseteq E$$

for every z.

Conversely, we show that the interior of E is contained in every C_z. Take (x, y) with

$$x^2 + 4y^2 < 1.$$

Choose $\alpha \in (0, 1)$ sufficiently close to 1 so that

$$\left(\frac{x}{\alpha}\right)^2 + 4\left(\frac{y}{\alpha}\right)^2 \leq 1.$$

Then

$$\left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) \in C.$$

Also, since C contains the z-axis,

$$\left(0, 0, \frac{z}{1-\alpha}\right) \in C.$$

By convexity, their convex combination also belongs to C, and

$$\alpha\left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) + (1-\alpha)\left(0, 0, \frac{z}{1-\alpha}\right) = (x, y, z).$$

Hence $(x, y) \in C_z$.

Thus, for every z,

$$\operatorname{int}(E) \subseteq C_z \subseteq E.$$

Since the boundary of the ellipse has area 0, we have

$$\operatorname{area}(C_z) = \operatorname{area}(E) = \frac{\pi}{2}$$

for every z. Therefore, the volume of C in the region $-1 \leq z \leq 1$ is

$$\int_{-1}^{1} \operatorname{area}(C_z)\,\mathrm{d}z = \int_{-1}^{1} \frac{\pi}{2}\,\mathrm{d}z = \pi.$$

Therefore, the answer is $\boxed{(2)\ \pi}$.

$\square$

Problem 2025.1. A fair coin is tossed repeatedly until either two heads in a row or two tails in a row appear. On average, how many tosses are needed before the process stops?

Solution. On the first toss, it is impossible to have two consecutive equal outcomes. After the first toss, the process stops exactly when the next toss gives the same result as the previous toss.

Since the coin is fair, from the second toss onward we have

$$\mathbb{P}(\text{the new toss matches the previous toss}) = \frac{1}{2}.$$

Therefore, after the first toss, the number of additional tosses needed until the process stops follows a geometric distribution with success probability $\frac{1}{2}$. Its expected value is

$$\frac{1}{1/2} = 2.$$

Since the first toss is always made, the expected total number of tosses is

$$1 + 2 = 3.$$

Thus the answer is $\boxed{3}$.

□

Problem 2025.2. The nineteenth-century mathematicians Cauchy and Weierstrass established rigorous definitions of limits. Who was the first mathematician to prove that the sum of the reciprocals of the squares of all positive integers is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}?$$

(1) David Hilbert
(2) Georg Cantor
(3) Bernhard Riemann
(4) Leonhard Euler

Solution. The problem of finding the exact value of the infinite series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is known as the Basel problem.

This problem had been known since the seventeenth century, and it was Leonhard Euler who first found and proved the famous identity

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore, the answer is $\boxed{\text{(4) Leonhard Euler}}$.

□

Problem 2025.3. Find the equation of the smallest sphere tangent to all four planes

$$x = 0, \qquad y = 0, \qquad z = 0, \qquad x + y + z = 3.$$

Solution. Since the sphere is tangent to the three coordinate planes

$$x = 0, \qquad y = 0, \qquad z = 0,$$

and we are looking for the smallest such sphere, its center must lie in the first octant and have the form

$$\mathbf{c} = (a, a, a)$$

by symmetry. The distance from $\mathbf{c}$ to each coordinate plane is a, so the radius of the sphere is

$$r = a.$$

The sphere must also be tangent to the plane

$$x + y + z = 3.$$

The distance from (a, a, a) to this plane is

$$\frac{|a + a + a - 3|}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{3 - 3a}{\sqrt{3}},$$

since the smallest sphere lies between the coordinate planes and the plane $x + y + z = 3$. This distance must equal the radius a, so

$$\frac{3 - 3a}{\sqrt{3}} = a.$$

Hence

$$3 - 3a = \sqrt{3}a,$$

and therefore

$$a = \frac{3}{3 + \sqrt{3}} = \frac{3 - \sqrt{3}}{2}.$$

Thus the center is

$$\left(\frac{3 - \sqrt{3}}{2}, \frac{3 - \sqrt{3}}{2}, \frac{3 - \sqrt{3}}{2}\right),$$

and the radius is

$$\frac{3 - \sqrt{3}}{2}.$$

Therefore, the equation of the sphere is

$$\boxed{\left(x - \frac{3 - \sqrt{3}}{2}\right)^2 + \left(y - \frac{3 - \sqrt{3}}{2}\right)^2 + \left(z - \frac{3 - \sqrt{3}}{2}\right)^2 = \left(\frac{3 - \sqrt{3}}{2}\right)^2}.$$

$\square$

Part V

Expected Problems